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Household clustering and seasonal genetic variation of *Plasmodium falciparum* at the community-level in The Gambia

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eLife Assessment

This article presents a **valuable** genetic spatio-temporal analysis of malaria-infected individuals from four villages in a highly seasonal transmission setting in The Gambia, covering the period between December 2014 and May 2017. Evidence generated by the study's laboratory and data processing approaches is **solid** and helps to advance the understanding of malaria in The Gambia, particularly due to its longitudinal design and the inclusion of asymptomatic cases.

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Abstract

Understanding the genetic diversity and transmission dynamics of *Plasmodium falciparum*, the causative agent of malaria, is crucial for effective control and elimination efforts. In some endemic regions, malaria is highly seasonal with no or little transmission during up to 8 months, yet little is known about how seasonality affects the parasite population genetics. Here we conducted a longitudinal study over 2.5 years on 1516 participants in the Upper River Region of The Gambia. With 425 *P. falciparum* genetic barcodes genotyped from asymptomatic infections, we developed an identity by descent (IBD) based pipeline and validated its accuracy against 199 parasite genomes sequenced from the same isolates. Genetic relatedness between isolates revealed a very low inbreeding level, suggesting continuous recombination among parasites rather than the dominance of specific strains. However, isolates from the same household were six-fold more likely to be genetically related compared to those from other villages, suggesting close transmission links within households. Seasonal variation also influenced parasite genetics, with most differentiation occurring during the transition from the low transmission season to the subsequent high transmission season. Yet chronic infections presented exceptions, including one individual who had a continuous infection by the same parasite genotype for at least 18 months. Our findings highlight the burden of asymptomatic chronic malaria carriers and the importance of characterising the parasite genetic population at the community-level. Most importantly, 'reactive' approaches for malaria elimination should not be limited to acute malaria cases but be broadened to households of asymptomatic carriers.

Author Summary

Malaria, caused by the parasite *Plasmodium falciparum*, remains a significant health challenge, particularly in regions like The Gambia where transmission is seasonal. Our study explored how seasonal changes impact the genetic diversity and transmission patterns of *P. falciparum* within communities. By tracking 1516 participants over 2.5 years and analyzing the genetic material of the parasites they carried, we discovered that malaria parasites in this region exhibit high genetic diversity, with frequent recombination events rather than dominance by specific strains. Interestingly, we found that parasites from individuals within the same household were much more likely to be genetically related, suggesting close transmission links within households. Seasonal shifts also influenced genetic relationships, with parasite populations becoming more diverse during peak transmission periods. Notably, one individual carried the same parasite strain for over a year without showing symptoms, highlighting the hidden reservoir of chronic malaria infections. Our findings suggest that malaria control strategies should extend beyond treating symptomatic cases to include household members of infected individuals, aiming to disrupt the wider transmission network within communities. This approach could be crucial for advancing malaria elimination efforts in regions with similar transmission dynamics.

Introduction

Malaria control and elimination efforts require a detailed understanding of *Plasmodium falciparum* population dynamics, particularly in regions with seasonal transmission. In The Gambia, substantial progress has been made in reducing malaria prevalence in the first decade of 2000. However, the decline has fallen short of meeting the 75% reduction target set for 2025, emphasizing the need for more targeted interventions[1]. In regions characterised by seasonal transmission, malaria clinical cases peak towards the end of the wet season, a pattern mirrored in The Gambia's transmission cycle. Its climate is characterised by a rainy season from late June to September, followed by a ~8 month-long dry season, typically without rainfall. Malaria cases are almost exclusively reported from September to December (the high transmission season), while malaria is virtually absent for the remaining 8 months (the low transmission season) [2]. During the high transmission season, malaria disproportionately affects rural, low-income populations in the eastern regions [3–6]. Understanding how the *Plasmodium falciparum* parasite population adapts to these seasonal variations is critical for informing targeted malaria elimination strategies.

Evidence suggests that imported cases play a minor role in maintaining malaria transmission, with the resurgence of malaria attributed to a persistent reservoir of asymptomatic chronic carriers that bridge two transmission seasons separated by several months of low to no transmission [7–10]. Previous studies in The Gambia and neighbouring Senegal demonstrated that individuals can harbor persistent infections across transmission seasons [6,11], with some infections maintaining transmissible gametocytes [12,13]. This reservoir of asymptomatic infections represents a major challenge to malaria control programs Various molecular methods and metrics are now available to characterise the parasite genetic diversity. At the population level, the average COI—the number of distinct parasite genotypes within a host—is often used to infer transmission intensity [14–18]. Genetic relatedness is assessed through Whole Genome Sequencing (WGS) or the cost-effective molecular barcode genotyping, the latter able to identify a unique parasite strain with as few as 24 loci [19]. From pairwise distances between such barcodes, it has been shown that relatedness tends to decrease with time and distance of the sampled infections both at the national level in The Gambia and at the local level in neighbouring Senegal [18,20]. Population-level genetic analyses, particularly using

Identity-by-Descent (IBD), provide a powerful tool for exploring parasite relatedness and recent recombination events. While Fst-based studies are valuable for long-term and large-scale comparisons, IBD allows for the resolution of recent recombination events, shedding light on fine-scale spatio-temporal transmission dynamics [21–23]. Extensive genomic epidemiology studies to date focused on clinical cases, yet such symptomatic cases only represent a minority of all *P. falciparum* infections [24,25]. Thus, relying exclusively on clinical cases may lead to miss a substantial fraction of the actual parasite population diversity.

In this study, we analysed *P. falciparum* genetic diversity in four nearby villages of The Gambia's Upper River Region through a longitudinal study spanning 2.5 years [11,26]. Using a combination of molecular genotyping and whole genome sequencing, we constructed a high-quality pipeline to assess COI and genetic relatedness at the community level. We aimed to elucidate how seasonal transmission cycles shape parasite population structure, including the stability of COI, the spatio-temporal patterns of IBD, the persistence of drug resistance alleles, and the contribution of asymptomatic chronic carriers to sustained transmission. These findings provide insights into the dynamics of parasite recombination and the genetic adaptations required for persistence in seasonal transmission settings.

Materials and methods

Study design and participants

Starting in December 2014, we recruited all residents from two villages (Madina Samako and Njayel, identified respectively with the letters 'K' and 'J'), with two additional villages (Sendebu and Karandaba, identified respectively with the letters 'P' and 'N') recruited from July 2016, all four villages being in the Upper River Region in The Gambia within 5 km of each other. Active case detection was conducted a total of 11 times over the 2-year period, with each sampling session occurring within a ten-day window. Symptomatic cases that occurred between September and December 2016 were also sampled. More information about the recruited participants can be found in a previous study [26]. In December 2016, 74 asymptomatic *P. falciparum* carriers were sampled monthly for up to 6 months until May 2017, with 42 of them being part of a cohort previously reported [11]. The study protocol was reviewed and approved by the Gambia Government/MRC Joint Ethics Committee (SCC 1476, SCC 1318, L2015.50) and by the London School of Hygiene & Tropical Medicine Ethics Committee (Ref. 10982).

Sampling and molecular detection of parasites

Details on the *P. falciparum* detection is provided in previous works [11,26]. Briefly, fingerprick blood samples were tested for *P. falciparum* by varATS qPCR. From July 2016 onwards, individuals testing positive for *P. falciparum* were invited to provide an additional 5 to 8 mL venous blood sample that was leucodepleted with cellulose-based columns (MN2100ff) and frozen immediately, as described by MalariaGEN (*https://www.malariagen.net/article/online-protocol-leucocyte-depletion-using-mn2100ff-cellulose-columns/*). DNA was extracted with QIAgen Miniprep kit following manufacturer procedure.

Genotyping and genome sequencing

A total of 522 *P. falciparum* DNA positive samples, 307 from fingerprick and 215 from venous blood, were processed for genotyping (442 samples) and whole genome sequencing when sufficient parasite DNA was available (331 samples of which 251 were genotyped) as part of the SpotMalaria consortium (**Figure 1B**). Genotyping was performed by mass-spectrometry based platform from Agena MassArray system. The output consisted of 101 bi-allelic SNPs located on the 14 chromosomes and concatenated into a 'molecular barcode' [27], plus six markers of resistance to antimalarials: AAT1 S528L (associated with chloroquine resistance), CRT K76T (associated with chloroquine resistance), DHFR S108N (associated with pyrimethamine resistance), DHPS A437G (associated with sulfadoxine resistance), Kelch13 C580Y (associated with artemisinin resistance) and MDR1 N86Y (associated with chloroquine, amodiaquine, lumefantrine and mefloquine resistance). The 101 SNPs had been picked for their variable allele frequencies within the *P. falciparum* population [27]. Whole genome sequencing (Illumina) was performed after a Selective Whole Genome Amplification step [28]. Paired end DNA sequence-reads (150 bp) were aligned to 3D7 reference genome version 3. Variants were called by a script from the MalariaGen consortium using GATK HaplotypeCaller [29,30]. Two drug resistance markers, AAT1 S528L and Kelch13 C580Y were only available in sequencing data and absent from genotyping data. If distinct calls of drug resistance markers were obtained between genotyping and whole genome sequencing data, only the call of whole genome sequencing was considered for the rest of the analysis.

Parasite relatedness

To accurately assess the parasite genetic similarity between different sampled infections, we estimated pairwise mean posterior probabilities of Identity-By-Descent (IBD) between genomes or barcodes using hmmIBD, a hidden Markov model-based software relying on meiotic recombination events given a recombination rate of *Plasmodium falciparum* of 13.5 kb/cM [31,32]. To estimate IBD, the model requires pairs of homozygous sites between two samples that are referred to as 'comparable sites' in this manuscript. These comparable sites are used by hmmIBD to yield the probability of two samples to be in IBD, which is equivalent to the expected shared fraction of their genomes. With an IBD of more than 0.9, two samples are considered identical, hence describing the same parasite genotype.

When samples are highly polyclonal, their set of homozygous sites tend to be made out of mostly alleles nearly fixed in the population, with a resulting IBD artificially high. Setting a minimal number of comparable sites for which at least one sample possess the minor allele will reduce this bias; these sites will be referred to as 'informative sites' in this manuscript.

Multi-locus genotype barcode data analysis pipeline

We developed a pipeline in R (version 4.3.3) combining Whole Genome Sequencing (WGS) and molecular genotyping data to produce a genetic relatedness network of parasites.

To analyse WGS data formatted in a VCF format, we developed the following pipeline to (**Figure S1**):

1. Filter out genomic loci for which the population minor allele frequency is inferior to 0.01, with more than 2 alleles identified in the population or that are located outside of the core genome [32]. Out of 1,042,186 initial SNPs, 27,577 (minimal QUAL of 132) were retained.
2. Remove genomes comprising less than 3000 SNPs covered by at least 5 reads.
3. Estimate the proportion of polyclonal samples using the F_{WS} metric and heterozygous loci.

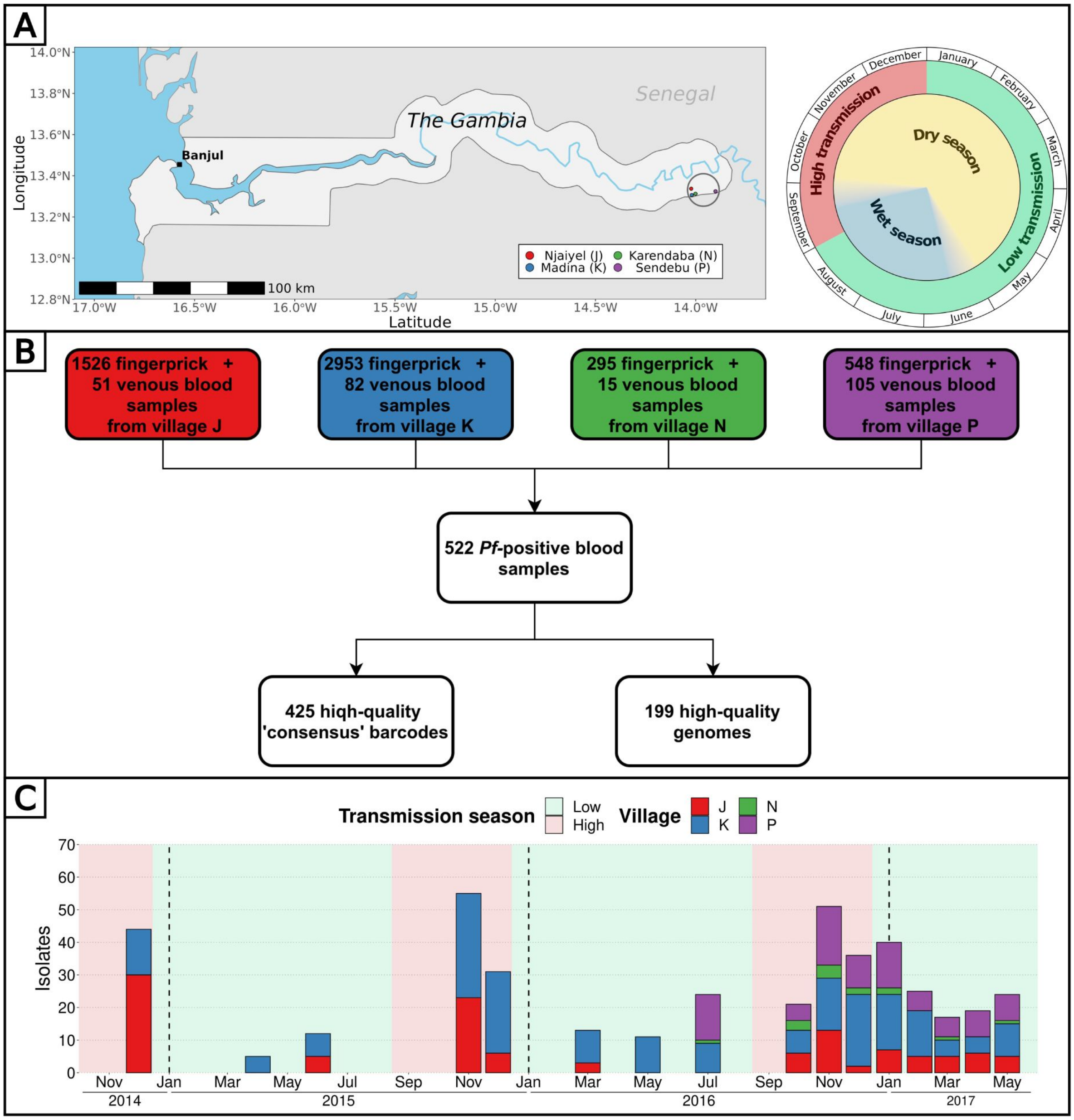


Figure 1

Study design and analysis pipeline.

(A) Blood samples from all participants from 4 villages of the Upper River Region of The Gambia were collected up to 16 times over 2.5 years. The peak of clinical malaria cases occurs at the end and right after the rainy season. Made with Natural Earth. (B) Overall, 522 blood samples (307 fingerprick and 215 venous blood) were genotyped and/or whole genome sequenced, resulting in 425 high-quality barcodes and 199 high-quality genomes. Additionally, 6 drug resistance markers were successfully genotyped and/or called from whole genomes, in a total of 438 isolates. (C) High-quality barcodes and genomes were sampled in 4 villages over 16 time points between December 2014 and May 2017.

4. Format SNPs into a matrix as required by hmmIBD. The matrix was formatted as follow: 0 and 1 were used for the respective allele of each SNP and -1 for mixed and unknown positions (which were then ignored during the IBD estimation). SNP calls were considered mixed if the within-sample Minor Allele Frequency (MAF) was greater than 0.2. The MAF of 0.2 was chosen according to the good agreement between molecular barcodes and genomic barcodes (high number of heterozygous call matches and low number of heterozygous-homozygous call mismatches) (**Figure S2**).
5. Use the paired IBD values obtained from hmmIBD and build a network of genome relatedness. IBD values were considered unknown between pairs of genomes having less than 100 informative sites (all our pairs had at least 1000 comparable sites).

The second part of the pipeline imputes missing SNPs in the molecular barcode from the WGS data to build a 'consensus barcode' (**Figure S1**):

1. Build a 'molecular barcode' out of the initial 101 genotyped SNPs and a 'genomic barcode' out of the same 101 SNPs called from high-quality genomes.
2. Remove 12 loci that are absent from all high-quality genomes.
3. Estimate the within-sample Minor Allele Frequency (MAF) to use as a cutoff to consider a locus homozygous when building barcodes out of genomic data. Genomic barcodes are built using different cutoffs of MAF below which a locus is considered homozygous and aligned against molecular barcodes from the same isolates. The cutoff of within-sample MAF of 0.2 showed a high number of heterozygous call matches and low number of heterozygous-homozygous call mismatches, meaning that molecular barcodes and genomic barcodes were in good agreement (**Figure S2**). This cutoff was retained to build all genomic barcodes. Each pair of molecular and genomic barcodes obtained from the same isolates were aligned. For isolates sampled after May 2016, molecular barcodes are most of the time not matching genomic barcodes for 21 loci (**Figure S3**). These 21 loci were clustered in one the 4 multiplexes used for molecular genotyping that probably failed to give the proper base calling. As a result, these 21 loci were considered unknown for all the molecular barcodes obtained after May 2016.
4. Using the alignment between molecular and genomic barcodes from the same isolates, replace unknown and mismatched SNP of the molecular barcodes by the SNP of genomic barcodes. These improved molecular barcodes are referred to as 'consensus barcodes'.
5. Remove consensus barcodes with fewer than 30 SNPs.
6. Estimate the proportion of polyclonal samples using heterozygous loci (described in following section).
7. Format consensus barcodes into a matrix as required by hmmIBD. The matrix was formatted like with genome data.
8. Use the paired IBD values obtained from hmmIBD and build a network of barcode relatedness. IBD values were considered unknown between pairs of barcodes having less than 30 comparable sites and 10 informative sites.

Complexity of infections

The clonality of each isolate was estimated from whole genome sequenced samples by the F_{ws} metric based on allelic frequencies from genomic data [33]. Additionally, the complexity of infections was estimated by the proportion of heterozygous loci (polyclonal if the proportion is above 0.5 % of available sites) in both consensus barcodes and genomes using a within-sample MAF of 0.2.

Genetic relatedness between groups of multi-locus genotype barcodes

All ‘consensus barcodes’(hereafter referred to as ‘barcodes’) available after December 2016 were excluded from the genetic relatedness analysis as they were obtained exclusively from the 74 individuals selected for their asymptomatic carriage of *P. falciparum* [11]. IBD was used to classify barcode (‘barcode-IBD’) pairs as related (IBD ≥ 0.5) or unrelated (IBD < 0.5). This classification was compared to the one resulting from IBD calculated between pairs of genomes (‘genome-IBD’, considered as gold-standard) using Cohen’s kappa [34]. For pairs of isolates with a different classification between barcode-IBD and genome-IBD, only the genome-IBD classification was retained (57/364 falsely related pairs set as unrelated and 58/18626 falsely unrelated pairs set as related). Additionally, the genome-IBD was used to classify barcodes as related or unrelated when the barcode-IBD was not available (710 pairs).

Barcodes were grouped by their collection date (11 time points from December 2014 to December 2016), sampling location (households or villages) or collection date split by household groups (within the same household, two households of the same village or two households of different villages). Within the same group, the genetic relatedness was estimated by the proportion of related barcodes (IBD ≥ 0.5) over all possible pairs of barcodes. When comparing different groups, the genetic relatedness was estimated by the proportion of related barcodes over all pairs of barcodes of each group, excluding pairs involving barcodes from the same individual. We used a stringent criteria of filtering out pairs of collection dates split by household groups with less than 5 comparisons (25/192 removed pairs). All pairs of villages (10 pairs) and all pairs of dates (66 pairs) contained at least 10 comparisons.

Results

Combined barcode and whole genome analysis pipeline

Overall, 5322 fingerprick and 253 venous blood samples were collected from 1516 individuals aged 3 to 85 years in four nearby villages of the Upper River Region of The Gambia between December 2014 and May 2017, as detailed in previous studies (**Figure 1A**) [11,26]. To characterise the *P. falciparum* population genetic relatedness and identify the impact of antimalarial drugs, we attempted parasite genotyping on 442 isolates (Table S4) and whole genome sequencing on 331 isolates (Table S5) for a combined total of 522 unique isolates over 16 time points (**Figure 1B**). Whole parasite genomes were successfully sequenced for 199 highly covered (between 3756 and 27516 high quality SNPs) isolates distributed across all 16 time points (**Figure S4**). Through the concatenation of SNPs from molecular barcode genotyping and whole genome sequencing, we obtained a high-quality ‘consensus’ barcode for 425 isolates comprising on average 65 SNPs (median of 64 SNPs, range 30 to 89 SNPs) (**Figure 1C**, FigureS5). A detailed description of the pipeline is available (**Figure S1**).

Complexity of infection is stable across seasons

The complexity of infection (COI), defined as the number of unique genotypes/genomes within an infected individual, serves as an indicator of parasite strain diversity within the population. We estimated COI using F_{ws} values from genomes and the proportion of heterozygous loci from genomes and barcodes. Across these metrics, the proportions of polygenotype isolates were estimated as 40 % (ranging from 22 to 61 % between time points) using F_{ws} (80/199), 39 % (range: 22 to 57 %) using heterozygous loci of genomes (78/199) and 34 % (range: 18 to 62 %) using heterozygous loci of barcodes (145/425). As expected, the proportion of heterozygous loci showed a strong negative correlation with F_{ws} both for barcodes (R^2 = 0.83, p-value < 10^{-15}) and genomes (R^2 = 0.92, p-value < 10^{-15}), indicating its efficacy as a predictor of COI (**Figure S6**). Although the proportion of polygenotype isolates fluctuated between time points, no discernible trend was observed between low and high transmission season, suggesting a relatively stable complexity of infections in the population throughout the 2.5-year study duration (**Figure S7**). At the individual level, we previously showed that the COI was stable in polyclonal infections during the dry season [11].

Low *P. falciparum* inbreeding at the community-level

To determine the relatedness between isolates and compare malaria parasites from distinct geographical locations and distant times, Identity By Descent (IBD) was calculated pairwise. The reliability of consensus barcodes in identifying genetically related isolates was confirmed through a strong linear correlation between barcode-IBD and genome-IBD, particularly when both were above 0.5 ($R^2 = 0.77$, p-value < 10^{-15}), with this threshold chosen to distinguish between related (IBD ≥ 0.5) and unrelated (IBD < 0.5) samples for all 425 consensus barcodes (**Figure S8**). The classifications (related or unrelated) resulting from barcode-IBD values and genome-IBD values (considered as the gold-standard) were in a strong agreement (Cohen's kappa of 0.839) with high values of specificity (0.997), sensitivity (0.841) and precision (0.843).

The level of inbreeding in the parasite population was characterized from relatedness estimated between barcodes sampled up to December 2016, discarding subsequent samplings in 2017 as they concerned exclusively a cohort of 74 asymptomatic carriers. To avoid sampling bias, only one barcode per continuous infection was retained, resulting in 284 unique barcodes sampled between December 2014 and December 2016. Among 34,326 pairwise comparisons of barcodes, only 435 (1.3 %) demonstrated relatedness with an IBD above 0.5, indicating extensive outcrossing within the parasite population (**Figure 2A**). Out of the 284 barcodes, 73 (26 %) were identical (IBD ≥ 0.9) to a barcode from another individual, while 140 (49 %) were related (0.5 ≤ IBD < 0.9) to at least one other barcode (**Figure 2B**). Although clusters of genetically related barcodes may suggest a higher connection within villages (1.6 % of related barcodes) than between them (0.7 % of related barcodes), the average proportion of related isolates was not significantly different (Welch t-test value = 1.42, p-value = 0.24), indicating interconnectedness between villages.

Pattern of relatedness between infections is shaped by seasonality

To explore the spatio-temporal relationship between barcodes from distinct individuals, we took advantage of our frequent samplings between December 2014 and December 2016 at the community-level to calculate the percentage of related isolates across six temporal groups, ranging from 0 to 2 months apart to 16 to 24 months apart between sample collections (**Figure 3A**). Remarkably, at the temporal level, the average proportion of related barcodes within barcodes sampled less than two months apart is 4.7 %. This value is ten-fold higher than the proportion of related barcodes sampled more than 12 months apart equal to 0.30 % (Welch t-test value = 4.72, p-value < 10^{-4}). This indicates that recombination between *P. falciparum* isolates breaks down IBD with time.

Similarly, the pairwise IBD was computed for spatial groups: pairs of barcodes from the same household, different households within the same village, and different households across villages (**Figure 3B**). At the spatial level, barcodes sampled less than 2 months apart and from the same household were six times more related than those sampled from different villages (average proportions of 0.093 and 0.015, Welch t-test value = 3.30, p-value < 0.05). However, when barcodes were sampled more than two months apart, the correlation between genetic relatedness and sampling location disappeared. Altogether, the overall low inbreeding combined with the increased proportion of related isolates within the same household indicate a scenario in which the same infectious mosquito infected two or more individuals living together, or a direct transmission chain between two household members.

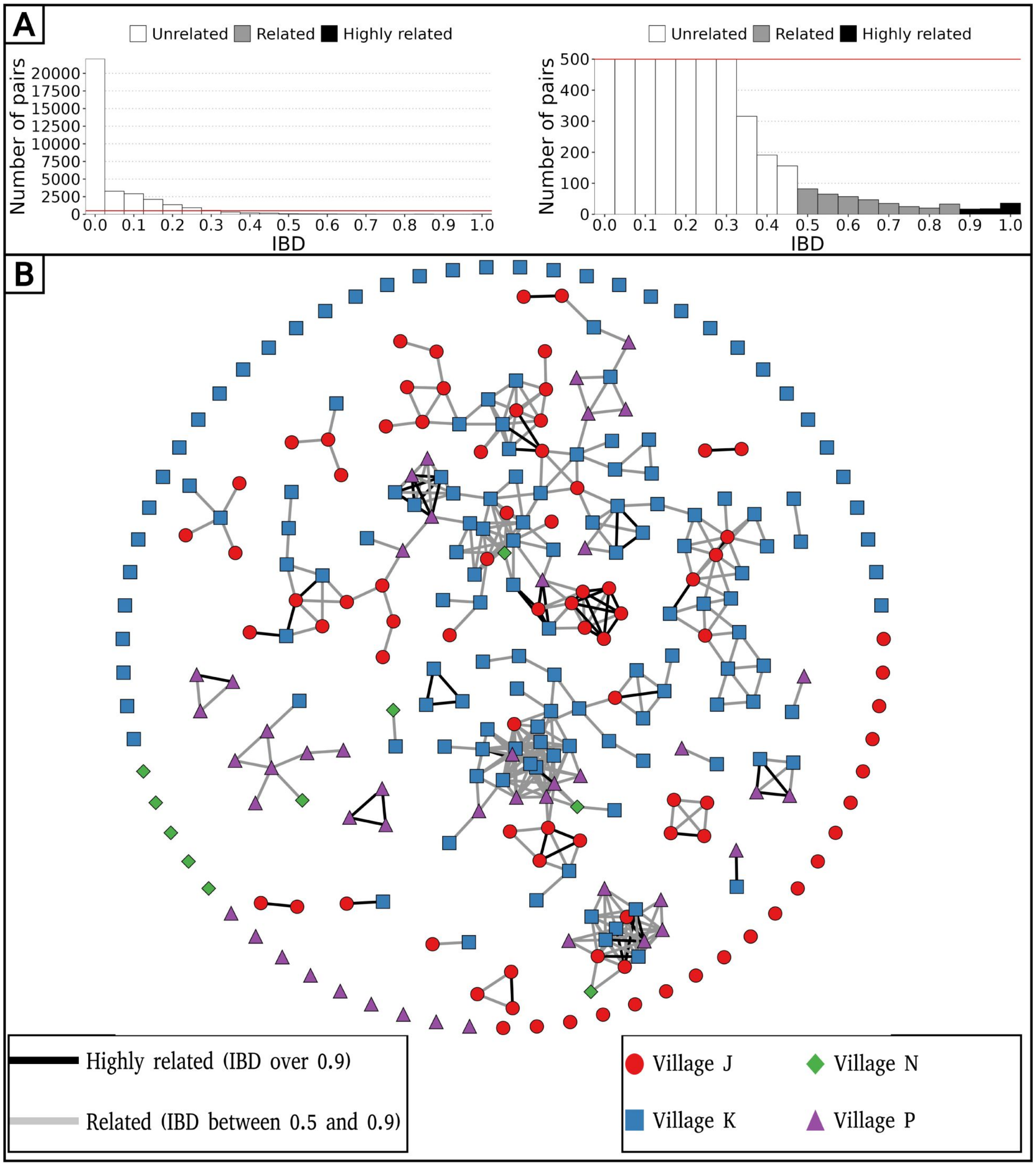


Figure 2

Low parasite inbreeding level in 4 villages in The Gambia inferred from inter-individual genetic relatedness.

The genetic relatedness of parasites was assessed from barcodes sampled between December 2014 and December 2016, keeping just one barcode per continuous infection, which resulted in 284 remaining barcodes. (A) Distribution of IBD values between barcodes (left panel), and with a cap at 500 pairs to highlight related barcodes at lower frequency (right panel). (B) Relatedness network of 284 isolates with barcodes represented as nodes and IBD values represented as edges. Barcodes are grouped into clusters using the compound spring embedder layout algorithm from Cytoscape (version 3.10.1).

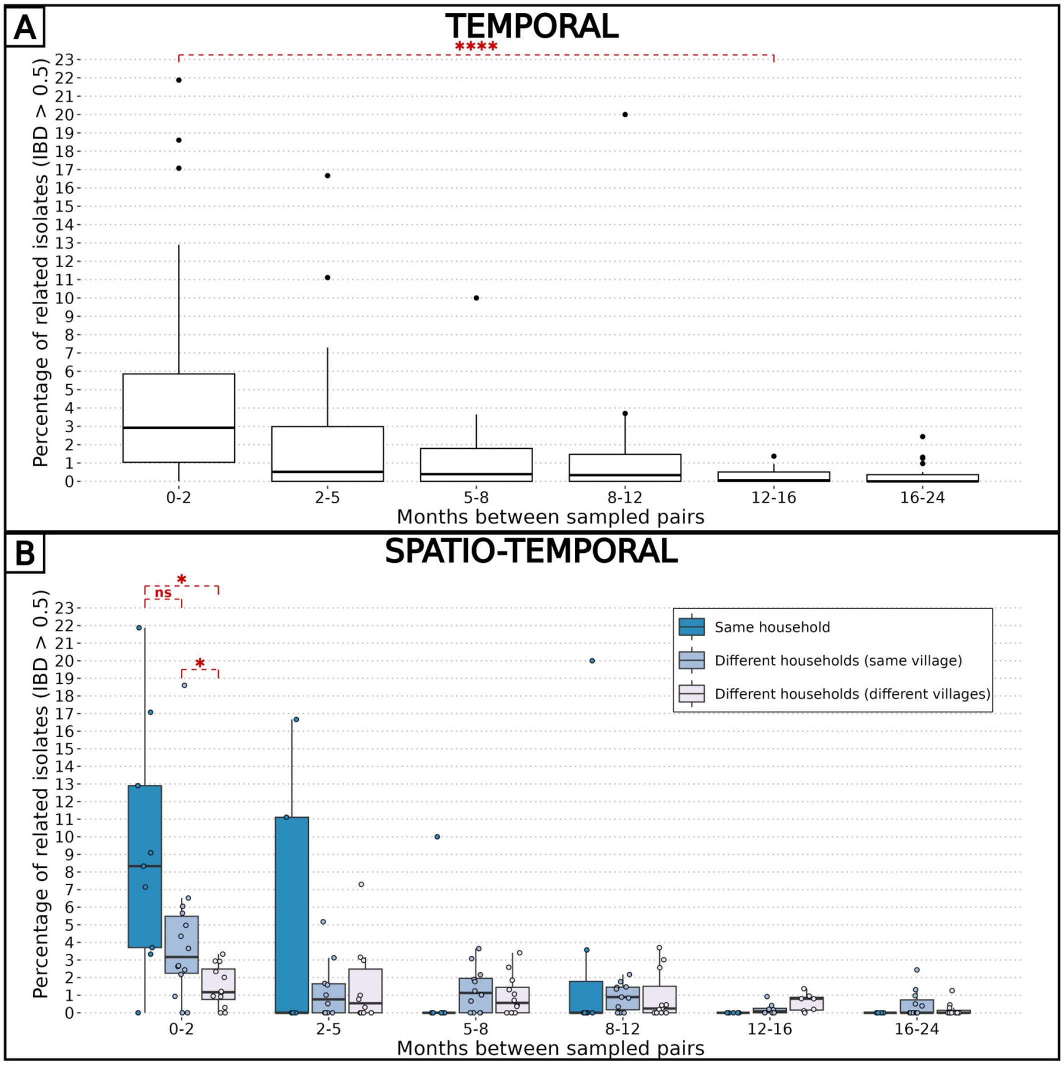


Figure 3

Combined effects of spatial and temporal distances on parasite relatedness.

The proportions of related barcodes (IBD ≥ 0.5) between each pair of households are binned into time intervals of various lengths such that the number of observations in each bin is similar. Each box shows the 25-75 % interquartile range of the percentage of related barcodes, with horizontal bar as the median. Groups of related barcodes were compared with Welch t-tests (*: p-value < 0.05, ****: p-value < 0.00005, ns: non-significant). (A) All pairs of isolates were grouped together in the same time interval. (B) Pairs of isolates were grouped by their relative spatial distance.

The impact of seasonality on the parasite inbreeding level was assessed by comparing the proportion of related barcodes (IBD ≥ 0.5) between groups of collection dates within or between transmission seasons. Barcodes sampled during the high transmission seasons of 2014, 2015 and 2016, as well as the low transmission seasons of 2015 and 2016, were grouped into intraseasonal pairs if they were collected during the same season or interseasonal pairs otherwise. This resulted in 5 groups for intraseasonal pairs and 10 groups for interseasonal pairs, with the most distant groups (high 2014 and high 2016) being 4 seasons apart (**Figure 4A**). As in **Figure 3**, pairs of collection dates close in time exhibited greater similarity than more distant collection dates. This suggests a continuous recombination process among all parasites, rather than the transmission of one or more specific strains. To determine the specific time of the year the average genetic relatedness declines, the proportion of related barcodes was compared between pairs of sample collections from the same season or one season apart (**Figure 4**). Parasites from the 'low to high' group exhibited the lowest average proportion of related barcodes (average relatedness of 0.006) compared with the 'within high' (average relatedness of 0.026, Welch t-test value = 3.14, p-value < 0.01), 'within low' (average relatedness of 0.024, Welch t-test value = 2.35, p-value < 0.05) and 'high to low' (average relatedness of 0.024, Welch t-test value = 4.87, p-value < 0.001) groups all displaying a 4-fold higher genetic relatedness. Here, most of the parasite differentiation occurs during the transition from the low transmission season to the subsequent high transmission season. This corresponds to the increase in transmission rate at the onset of the high transmission season, with parasite genomes being reshuffled after sexual reproduction in the mosquito. This decrease in relatedness is not observed in the 'high to low' group, demonstrating a reduced transmission in the low transmission season.

Independence of seasonality and drug resistance markers prevalence

In The Gambia, drug resistant allele frequencies increased dramatically from the late 1980's until the early 2000's, then plateaued until 2008 [35]. The rise in clinical malaria cases during the high transmission season implies usage of anti-malaria drugs, particularly among children aged between 0 and 5 years receiving antimalarials (sulfadoxine-pyrimethamine and amodiaquine) monthly during Seasonal Malaria Chemoprevention (SMC), from September to November [36]. The artemisinin-based combination therapy Coartem® (artemether-lumefantrine) is the first-line antimalarial used to treat uncomplicated malaria in The Gambia at the time of the study. We investigated whether the prevalence of drug resistance alleles is influenced by malaria seasonality due to differential selective pressures applied between high (more pressure) and low (less pressure) transmission seasons. To determine the prevalence of resistant haplotypes over time, six drug resistance-related haplotypes were obtained by both molecular genotyping and whole genome sequencing in genes *aat1*, *crt*, *dhfr*, *dhps*, *kelch13* and *mdr1*, leading to a merged total of 438 isolates with drug resistance-related haplotypes. Overall, 89 % of haplotypes called from both molecular genotyping and whole genome sequencing were identical (**Figure S9**).

As expected, the haplotype Kelch13 C580Y, related to artemisinin resistance, was absent. Overall, the proportion of isolates with resistant alleles was stable over time for AAT1 S258L (0.92, 95 % Wilson's Confidence Interval: 0.87-0.95), CRT K76T (0.64, 95 % CI: 0.59-0.70), DHFR S108N (0.93, 95 % CI: 0.90-0.95) and MDR1 N86Y (0.12, 95 % CI: 0.09-0.17) (**Figure 5**). In contrast, the haplotype DHPS A437G appears to decrease in prevalence twice, from the 2015 and 2016 high transmission seasons to the subsequent 2016 and 2017 low transmission seasons. As DHPS A437G haplotype has been associated with resistance to sulfadoxine, its apparent increase in prevalence during high transmission seasons could be resulting from the selective pressure imposed on parasites during SMC. Overall, our data from 2015 to 2017 is very similar to the allele frequency levels from 2008, indicating a potential plateau in the cost-benefit of drug resistance alleles [35].

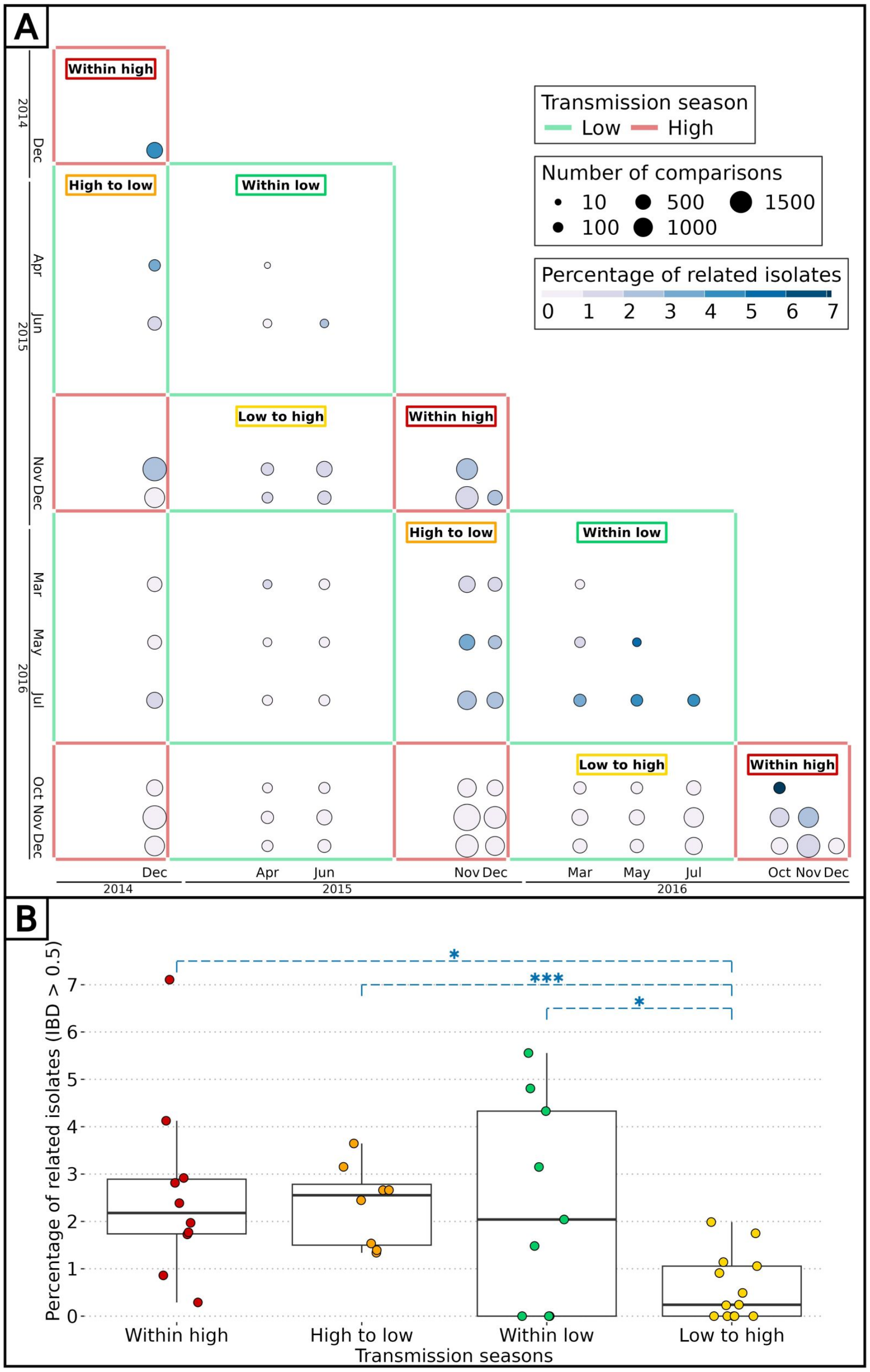


Figure 4

Effect of seasonality on parasite inbreeding level.

(A) Proportion of related barcodes (IBD ≥ 0.5) between all sample collections from December 2014 to December 2016. Each square represents a set of pairs of time points within the same seasonal pair. There were 5 successive transmission seasons during the study, hence 15 unique seasonal pairs. Annotations are present for seasonal pairs corresponding to a season compared with itself ('within high' and 'within low') and to a season compared with the preceding/succeeding one ('low to high' and 'high to low'). (B) Proportion of related barcodes between pairs of sample collections within the same season ('within high' and 'within low') and one season apart when the high transmission season precedes the low transmission season and conversely (respectively 'high to low' and 'low to high'). Genetic similarities were compared between the 'low to high' group and all other groups with Welch t-tests (*: p-value < 0.05, ***: p-value < 0.0005).

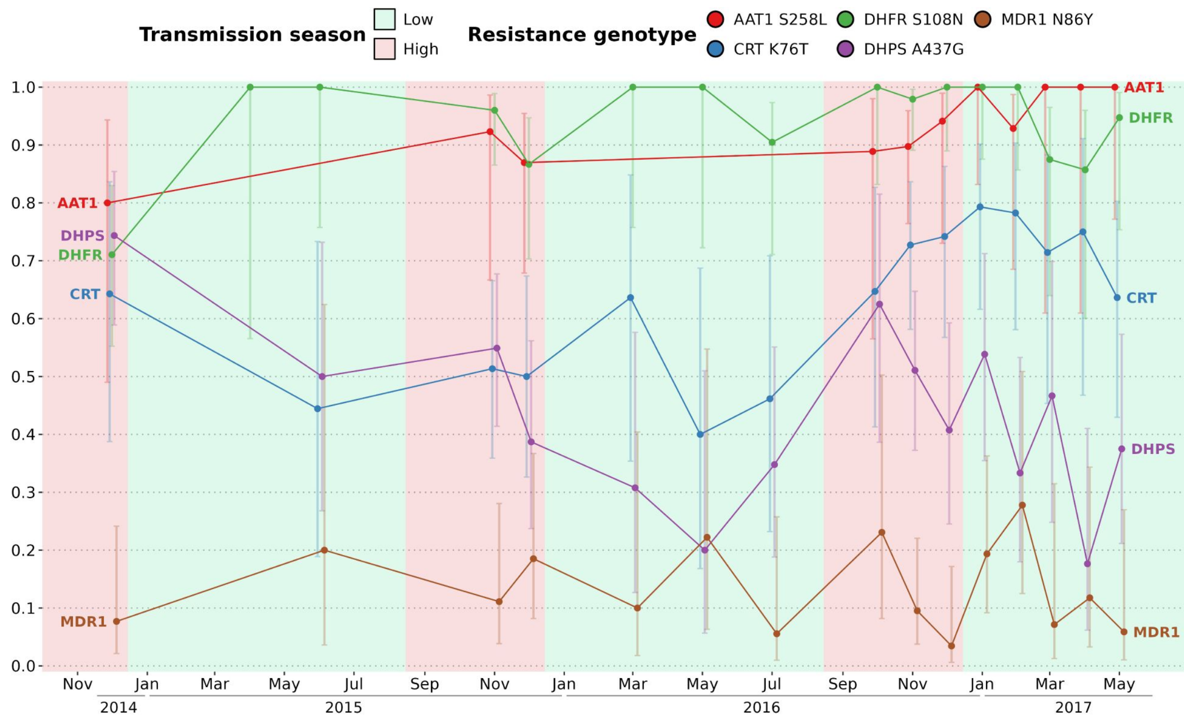


Figure 5

Prevalence of 5 drug resistance-related haplotypes for each time point with at least 10 observations over the study time period.

The 5 variants induce non-synonymous changes in AAT1 S528L, CRT K76T, DHFR S108N, DHPS A437G and MDR1 N86Y which are known to reduce the susceptibility to multiple antimalarials. The prevalence of the Kelch13 C580Y mutation was null and not shown here. For all markers except DHPS A437G, the proportions of all variants remained stable between the start and end of the study period. Error bars represent the 95 % Wilson's confidence intervals.

P. falciparum chronic infections with persisting genotypes

To distinguish reinfections from 'true' chronic infections with the same parasite genotype, we measured the minimal duration of infection using IBD values between barcodes obtained from the same individual sampled at different time points (which were not utilized for the spatio-temporal analysis of genetic relatedness). The beginning and end of an infection by the same parasite were attributed to the two most extreme time points separating two identical barcodes (IBD ≥ 0.9). Overall, 32 individuals (20 males and 12 females) were chronically infected with the same dominant *P. falciparum* genotype ranging from one month to one and a half years (**Figure 6**). Out of the 32 individuals chronically infected, 21 were asymptomatic carriers selected for a monthly sampling between December 2016 and May 2017, rising the chance of observing at least twice the same dominant *P. falciparum* genotype. Gender and age did not significantly influence the duration of infection (**Figure S10**). Two individuals (24 and 25) were infected in 2017 with two distinct parasite strains, as shown by monthly barcodes alternating between the two strains (for both individuals, barcodes from Feb and Apr, and from Mar and May, are identical). Interestingly, for individual 24, the two strains were genetically related (IBD ≈ 0.5), likely indicating co-infection of sibling strains from the same brood with a fluctuating relative abundance over time. The change in proportion between the two strains over time might be caused by intra-host competition (e.g. one strain, benefiting from immune evasion, is more abundant than the other) or asynchronous developmental stages (one strain is mostly sequestered when the other is mostly circulating and conversely).

Discussion

In anticipation of the pre-elimination phase of malaria, our study aimed to develop robust methods for estimating the complexity of infections (COI) and constructing genetic relatedness networks of *P. falciparum* parasites. By tracking 1,516 participants over 2.5 years, we found that malaria parasites in this region exhibit high genetic diversity, driven by frequent recombination events rather than dominance by specific strains. Notably, parasites from individuals within the same household were significantly more likely to be genetically related, indicating close transmission links within households. Additionally, seasonal changes influenced the genetic relationships among parasites, with populations becoming more diverse during peak transmission periods, reflecting the dynamic nature of malaria transmission in this setting.

Leveraging both barcodes and genomes offers complementary advantages: molecular barcode genotyping provides cost-effective access to genetic information from numerous blood isolates, while whole genomes yields quantitative data on thousands of single nucleotide polymorphisms (SNPs), facilitating comprehensive analyses of parasite diversity, with the added potential to extend analyses to newly discovered haplotypes unidentified at the time of sequencing. Combining both data types validates base calling from molecular genotyping and guides method applicability across the limited positions available in barcodes.

The stable COI over time reflects a constant transmission intensity [16–18,37]. Such stable transmission in this area of The Gambia, reported already elsewhere, contrasts with the overall decrease of malaria transmission across the country, highlighting the heterogeneity malaria within The Gambia [6,26]. Our findings underscore the importance of understanding local malaria dynamics amidst broader regional trends.

With Identity by Descent (IBD), we evaluated parasite relatedness between genomes and between barcodes. Although using more SNPs (e.g. more than 200 SNPs) is ideal to reach an accurate relatedness, *in silico* data suggests that the accuracy is only marginally improved with 96 SNPs or

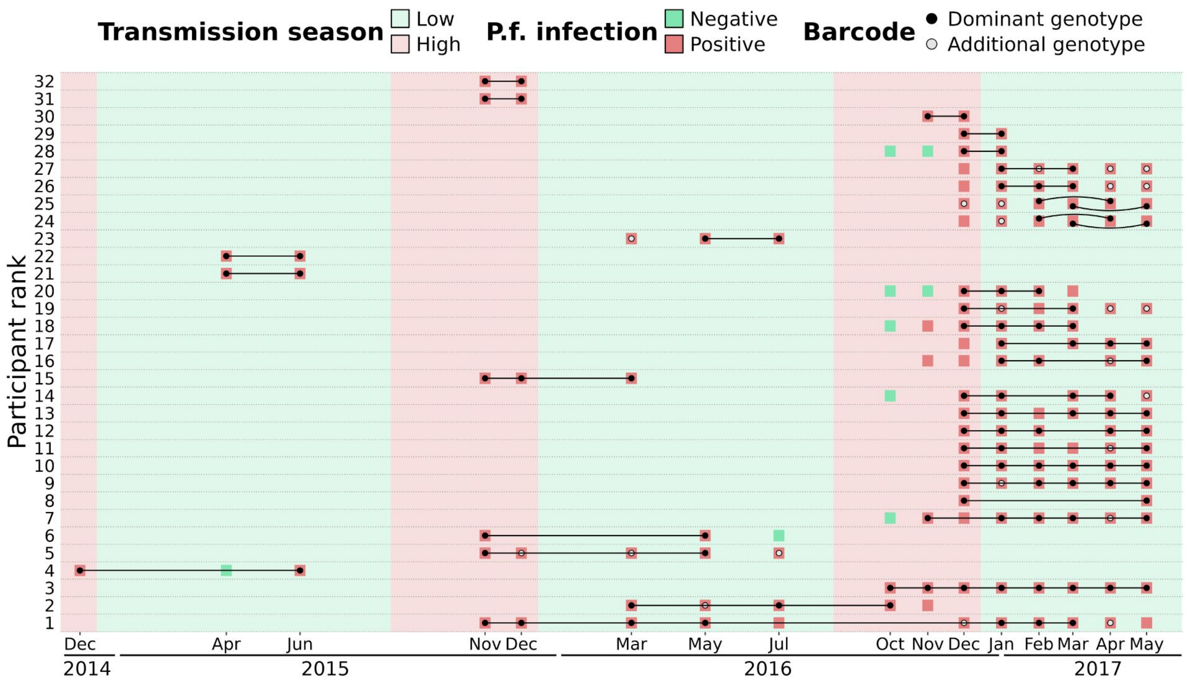


Figure 6

Continuous *P. falciparum* infections with the same dominant genotype.

A total of 32 individuals were infected with highly related barcodes (IBD ≥ 0.9) at two or more time points. Individuals are ranked by their duration of continuous infection from the longest to the shortest, with a black line linking identical 'dominant' genotypes (black dots) that are the farthest away from each other. Additional genotypes (grey dots) are different (IBD < 0.9) from the dominant genotype but may still be related (IBD ≥ 0.5). For two individuals (ranked 24th and 25th), two different parasite strains were present concurrently, represented by curved lines. Barcodes and *P. falciparum* infection status are shown up to 90 days prior or after the inferred continuous infection.

more [38]. Furthermore, we showed that for IBD values above 0.5, the correlation between genome-and barcode-IBD was very strong. Finally, the qualitative aspect of our approach, with isolates either related (IBD ≥ 0.5) or unrelated (IBD < 0.5) enables to compensate for the lower accuracy of barcode-IBD.

Overall, 26 % of unique parasite strains were found in two or more individuals and around 28 % of isolates were polyclonal according to the proportion of heterozygous loci of barcodes. This seemingly contrasts with findings from Thiès, in neighbouring Senegal, where parasite haplotype diversity is low, the proportion of shared genetic types between isolates is 35 %, with less than 10% of polyclonal infections, suggesting self-mating transmission and low outcrossing levels [9,10,39]. However, the key difference of our approach was the active detection of asymptomatic infections, as opposed to the large number of studies sequencing parasites collected from clinical cases. Our results argue for active case detection as a necessary step to comprehensively characterise the parasite genetic diversity.

One limitation of genetic epidemiology studies such as this one, is the necessity to exclude IBD values obtained between isolates with high level of polyclonality and thus not enough informative loci. One important remaining challenge is to incorporate highly polyclonal infections in the network of genetic relatedness of parasites, while the proportion of complex infections tends to increase with malaria transmission intensity [40]. Achieving this goal is difficult as it requires whole genome sequencing followed by a deconvolution tool such as DEploid, with the caveat that the different strains are not too related or too rare [41,42]. A second limitation to our study is that we cannot exclude that imported malaria cases might have an effect on our measured inbreeding levels, enough though previous findings tend to indicate a limited effect of imported cases on maintaining transmission [7–10]. Future studies will require recruiting a larger number of participants with more frequent samplings.

A follow-up study carried out in rural villages of eastern Gambia between 2012 and 2016 reported that roughly half of asymptomatic infected individuals at the end of the wet season were still infected at the end of the dry season [6]. We previously reported a similar rate [11]. We showed that parasites collected during the dry season share significantly more genetic similarity with those from the previous wet season than those from the next wet season. Average relatedness quickly declined with the temporal distance between samplings (10-fold lower after one year) probably due to active genetic recombinations in the population leading to fewer fragments in IBD over time. A study conducted in Colombia (with a 10-fold lower malaria prevalence than in The Gambia) showed that a 10-fold decrease in genetic similarity was reached in 9 years while we observed the same decrease after just one year in our study [37,43].

Most infections were sub-microscopic, which is typically observed when prevalence falls below 20 % [26,44,45]. These low-density infections that may persist for months are typically asymptomatic unless the host immune system is compromised [46]. In this study, the infection with the longest duration was in an asymptomatic individual aged between 5 and 9 years in which it persisted as the same parasite strain for one and a half year. School-age children contribute the most the malaria reservoir through their higher carriage of asymptomatic infections and their ability to effectively infect mosquitoes [6,12,47,48].

In the late 20th century, when the malaria burden in The Gambia was much higher, clinical cases from the same household were more likely to be caused by the same mosquito bite [49]. More recently, Ngwa *et al.* showed that parasite genetic similarity during the 2013 transmission season was inversely correlated with each spatial and temporal distances in The Gambia [20]. We also found that parasites sampled in neighbouring households are more genetically similar but only when sampled less than three months apart. Similarly, two other studies observed parasite strains with varying levels of spatio-temporal propagation in Thiès, Senegal, suggesting that some parasites are more actively transmitted than others [9,18]. These results add evidence that anti-malarial strategies should target all members of a household with an infected individual. Such “reactive” strategies include treatment of a malaria case and all its household members (without testing) or testing all household members and treating if necessary. The impact of reactive strategies on malaria transmission is at best very limited [50–53]. Based on the sheer size of the

P. falciparum asymptomatic reservoir, with parasitaemias typically below microscopy or RDT detection level [26], it is not surprising that targeting clinical cases and their immediate families is not sufficient to break the transmission. For a malaria elimination campaign, based on our insights and previous research [54] on asymptomatic infections in The Gambia at the community level, we argue for mass detection with a highly sensitive method such as qPCR, followed by treatment of *P. falciparum* positive cases and all their household members.

Data availability

The barcode data analysis pipeline can be found at *https://github.com/marcguery/malaria-barcodes-genomes-gambia*.

Supplementary tables are provided. This publication uses data from the MalariaGEN SpotMalaria project as described in ’Jacob CG et al.; Genetic surveillance in the Greater Mekong Subregion and South Asia to support malaria control and elimination; eLife 2021;10:e62997 DOI: 10.7554/eLife.62997’ [27]. The project is coordinated by the MalariaGEN Resource Centre with funding from Wellcome (206194, 090770).

Supplementary figures

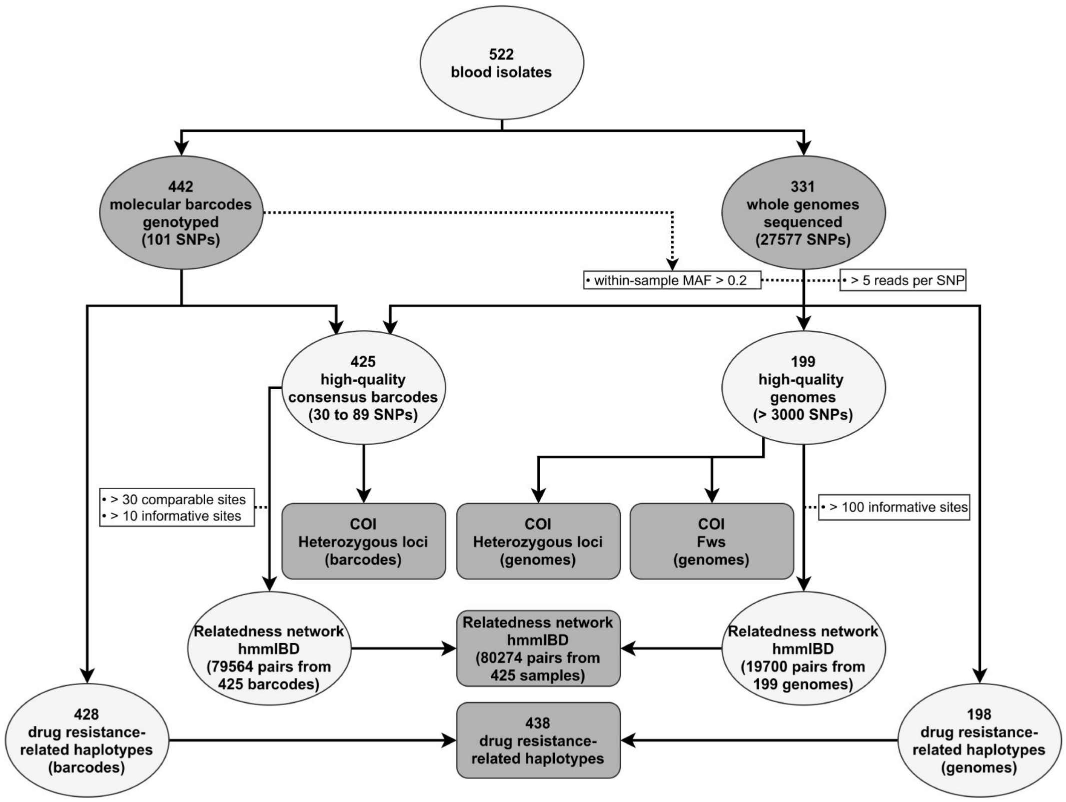


Figure S1

Steps of the combined barcode-and genome-analysis pipeline using 522 *P. falciparum*-positive blood isolates.

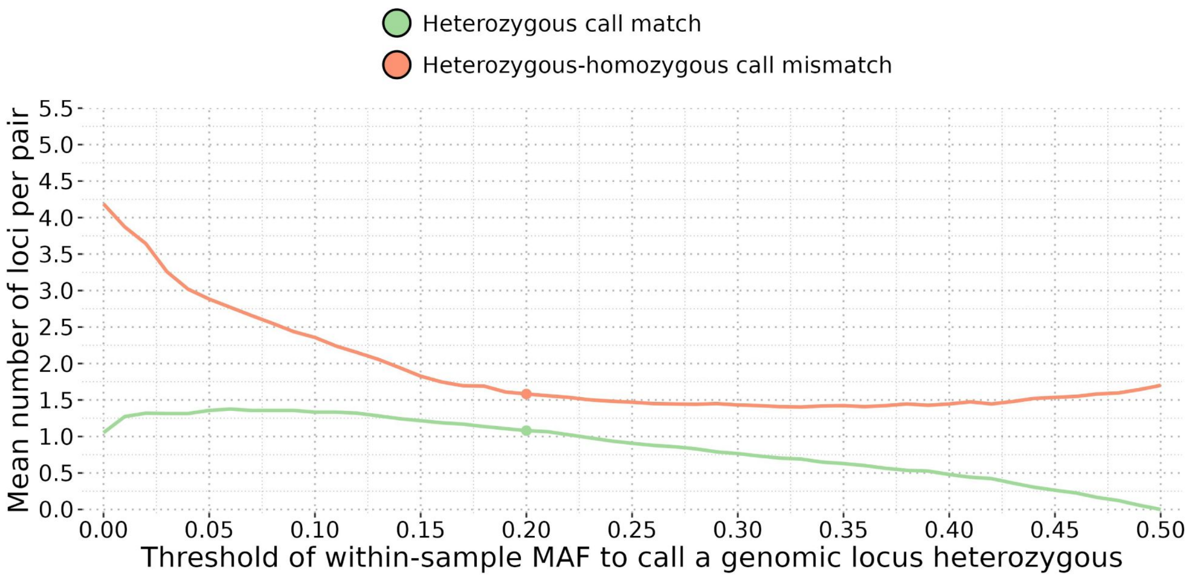


Figure S2

Comparison of heterozygous locus calls between each pair of molecular barcode (obtained by genotyping) and their corresponding genomic barcode (built from allelic frequencies).

Genome loci were considered heterozygous if their within-sample Minor Allele Frequencies (MAF) were above a threshold ranging from 0 to 0.5. The threshold of MAF of 0.2 was finally retained because it yields a high number of matches and a low number of mismatches of heterozygous locus call, making the genomic barcodes as close as possible to their molecular counterparts.

Figure S3

Comparison of molecular barcodes loci (built from genotyped SNPs) and consensus barcodes loci (combining molecular and WGS loci) with WGS loci for two groups of samples clustered by their collection dates.

(A) For each sample with both a barcode and a genome, pairs of corresponding loci are colour-coded as 'matching' (green), 'mismatching' (red), 'heterozygous call matching' (light green), 'heterozygous-homozygous call mismatching' (light red) or 'incomparable when at least one call is unknown' (grey). Barcode loci are sorted based on their agreement with WGS loci (proportion of matching loci over all comparable loci) from low to high.WGS loci were considered heterozygous if the within sample MAF was above 0.2 (Figure S2). (B) Average agreement between molecular barcode loci and WGS loci coloured by their average agreement in samples collected after May 2016 (< 0.8 coloured in red, ≥ 0.8 coloured in green). Barcode loci are sorted by their chromosomal location. Overall, loci match between the two methods for samples collected up to May 2016 (all loci have an agreement above 0.79). However, 21 genotyped loci show a consistent mismatch (all loci have an agreement below 0.7) with WGS loci for all samples collected after May 2016. These 21 genotyped loci were incorrectly genotyped because of a failure of one of the multiplexes during the genotyping procedure. To increase the number of available loci of barcodes, all unknown and incorrectly genotyped loci were replaced by WGS loci, resulting in a 'consensus barcode'.

Figure S4

Number of high-quality barcodes (425 isolates) and genomes (199 isolates) successfully sequenced over 16 time points between December 2014 and May 2017.

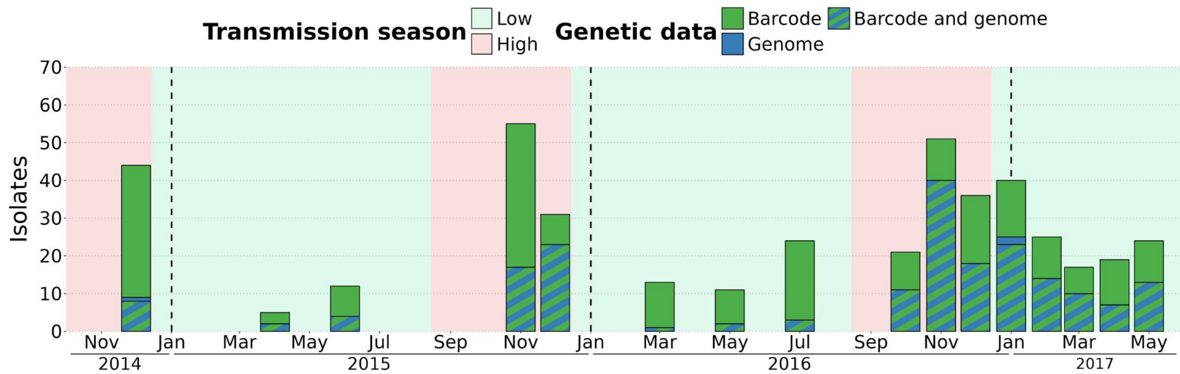
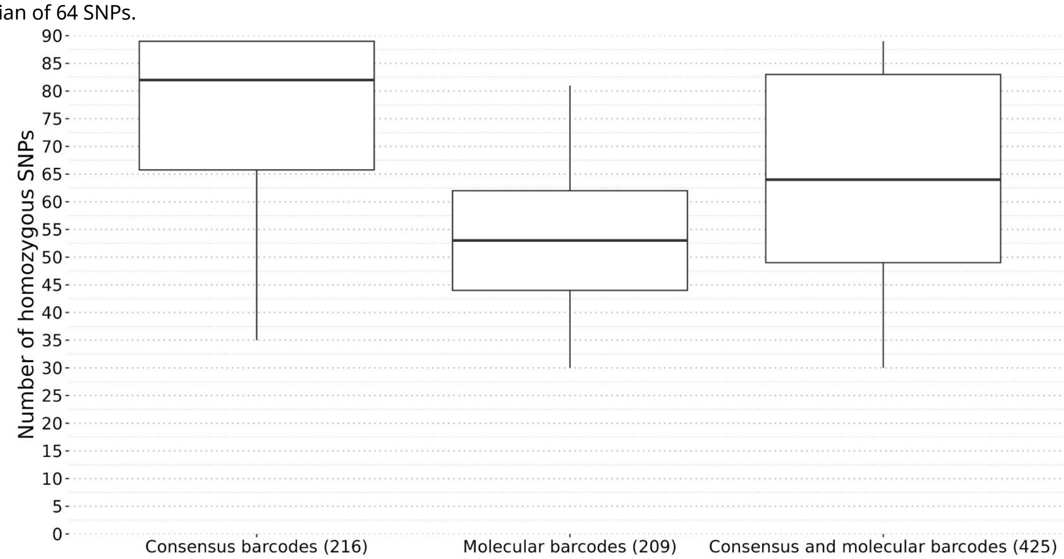


Figure S5

Number of homozygous SNPs per barcode.

Among 425 barcodes, 216 were improved using corresponding genomic calls (named 'consensus barcodes' on this figure), resulting in an increase in the number of homozygous SNPs (average: 76 SNPs, median: 82 SNPs). The remaining 209 molecular barcodes had a lower number of homozygous SNPs (average: 54 SNPs, median: 53 SNPs). Overall, the 425 consensus and molecular barcodes (referred to as 'consensus barcodes' in the main text) had an average of 65 SNPs and a median of 64 SNPs.



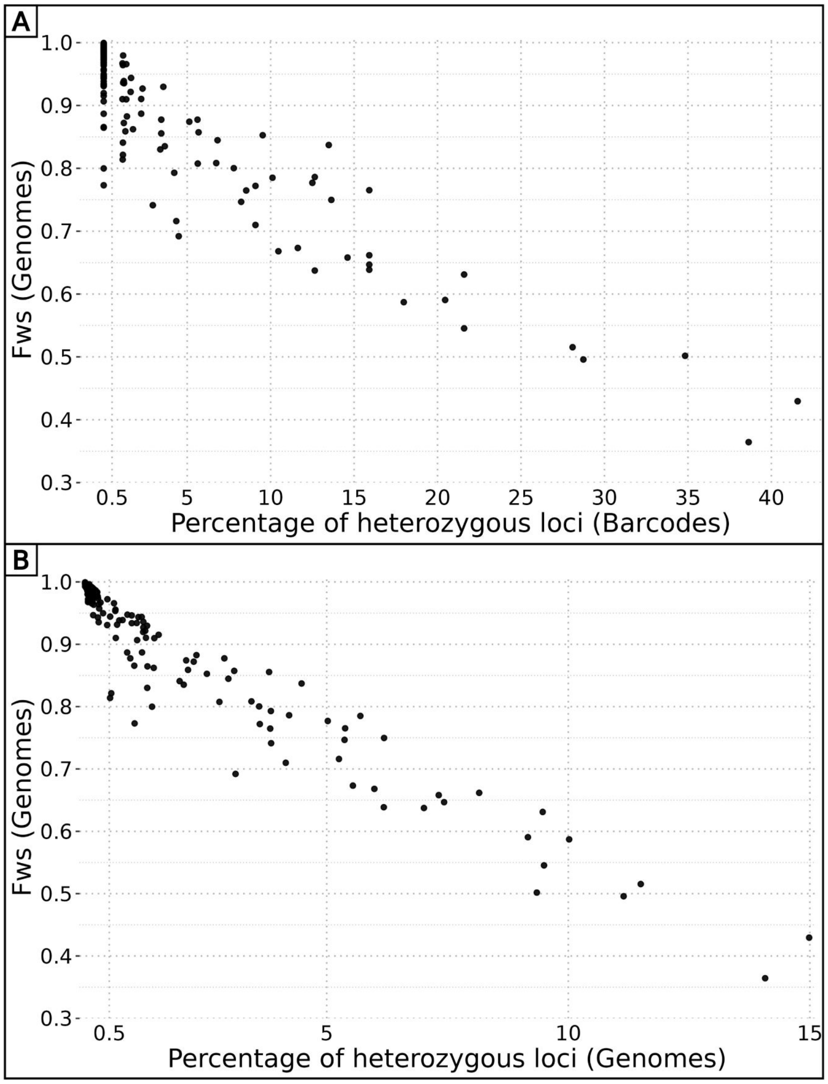


Figure S6

Correlation between F_{ws} values and percentage of heterozygous loci.

(A) The percentage of heterozygous loci in barcodes ranges from 0 to more than 40 %, highlighting their high variability among *P. falciparum* strains. The percentage of heterozygous loci shows a strong negative correlation with their corresponding F_{ws} value ($R^2 = 0.83$, p-value < 10^{-15}). Considering isolates with F_{ws} values available, only 15 % (20/132) of monoclonal isolates (heterozygous site proportion below 0.005) have a F_{ws} value below 0.95 and 6 % (4/64) of polyclonal isolates (heterozygous site proportion above 0.005) have a F_{ws} value above 0.95. (B) The percentage of heterozygous loci in genomes range from 0 to less than 15 %, which is expected as the 27,577 SNPs had various population-level MAF. The percentage of heterozygous loci shows a strong negative correlation with their corresponding F_{ws} value ($R^2 = 0.92$, p-value < 10^{-15}). Computing the proportion of heterozygous loci in both genome and barcode data (84 SNPs) enables to accurately estimate the level of clonality of an isolate.

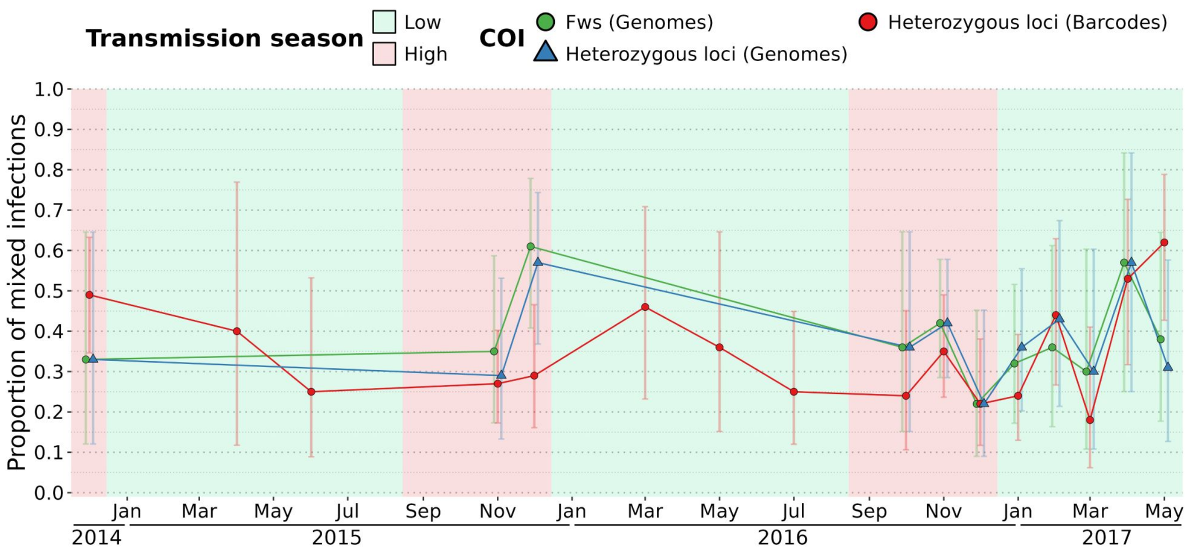


Figure S7

Proportion of polyclonal isolates over time estimated by F_{ws} (F_{ws} < 0.95) and the proportion of heterozygous loci (more than 0.5 % of available sites) on barcodes and genomes.

The percentage of polyclonal isolates is stable over time for all methods (around 40 % for F_{ws}, 39 % using heterozygous loci of genomes, 34 % using heterozygous loci of barcodes). Error bars represent the 95 % Wilson's confidence intervals.

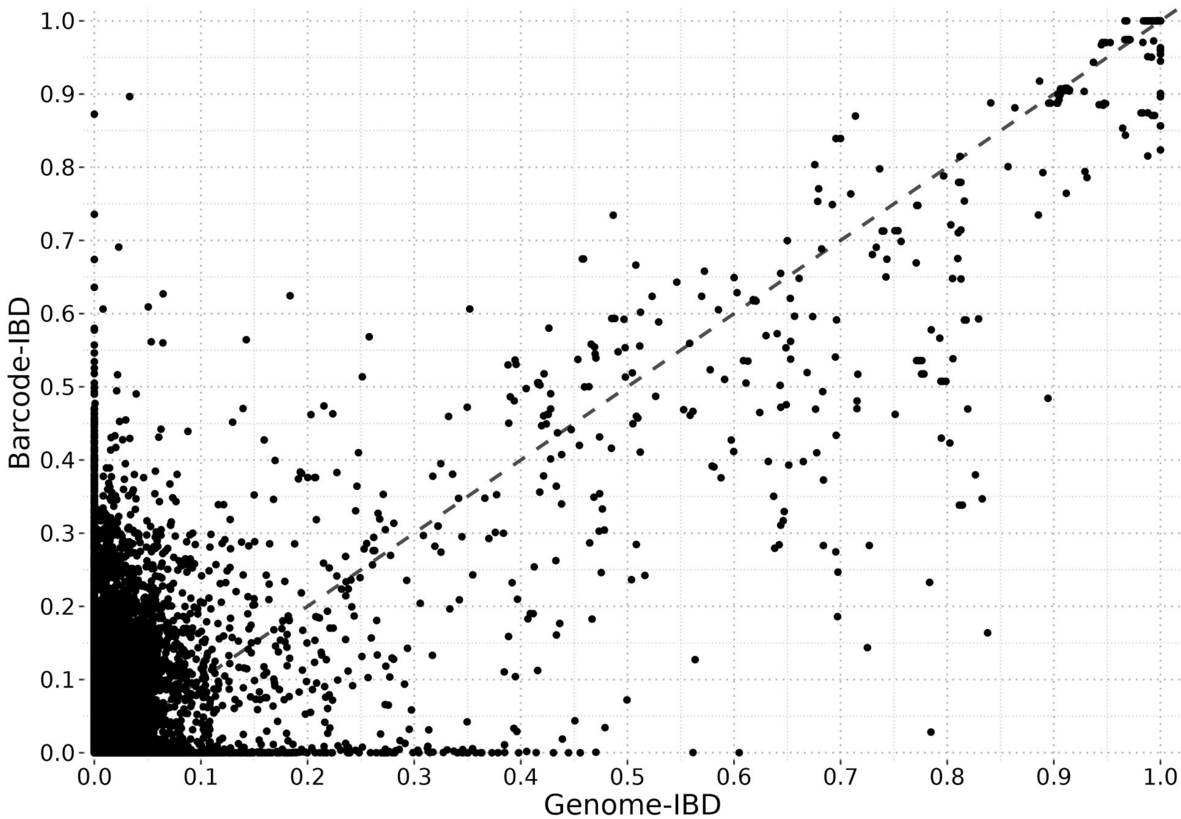


Figure S8

IBD calculated from consensus barcodes highly correlates with IBD from genomes.

Identity By Descent was calculated pairwise between the 425 high-quality consensus barcodes ('barcode-IBD') with at least 30 comparable sites and 10 informative positions (79564/90100 pairs remaining) and between the 199 high-quality genomes ('genome-IBD') with at least 100 informative positions (19700/19701 pairs remaining). The accuracy of the classification of related (IBD ≥ 0.5) and unrelated (IBD < 0.5) isolates from barcode-IBD was assessed using genome-IBD as the gold standard. Overall, the barcode-IBD classification was in a strong agreement with the genome-IBD classification (Cohen's kappa of 0.839) and had high values of specificity (0.997), sensitivity (0.841) and precision (0.843). Among the 364 pairs of samples with a barcode-IBD above 0.5, 314 were indeed related (genome-IBD above 0.5) while 50 were actually not related (genome-IBD below 0.5), although 24 showed genome-IBD value between 0.4 to 0.5. Regarding the 18626 unrelated pairs according to barcode-IBD classification, only 58 were called as related by the genome-IBD classification.

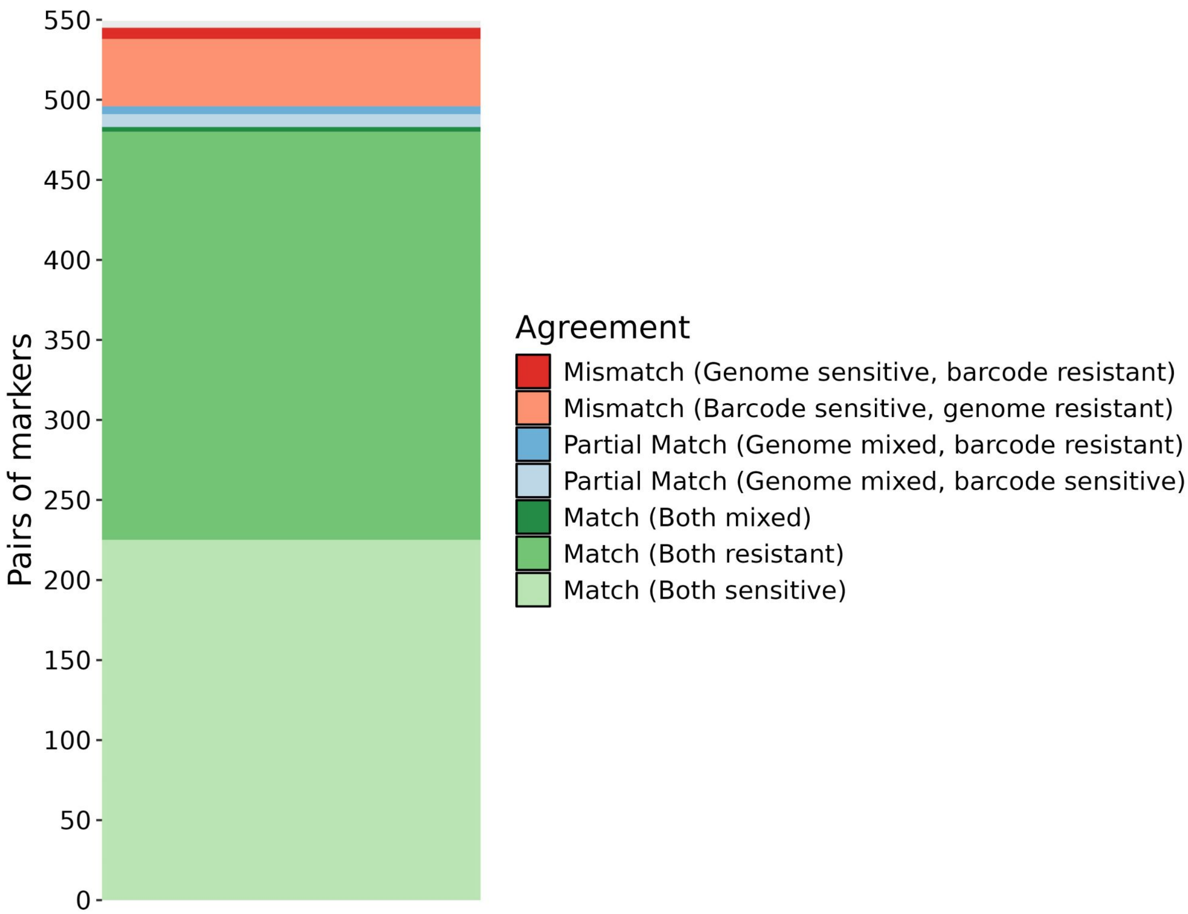


Figure S9

Agreement between pairs of haplotypes obtained both by molecular genotyping ('barcodes') and from Whole Genome Sequence calling ('genomes').

Six drug resistance markers were screened, AAT1 S528L, CRT K76T, DHFR S108N, DHPS A437G, Kelch13 C580Y and MDR1 N86Y. Drug resistance haplotypes were obtained in 428 barcodes (1261 haplotypes) and 198 genomes (1000 haplotypes) for a merged total of 438 isolates (1716 haplotypes). The vast majority (89 %) of the 545 pairs of haplotypes obtained with the two methods are identical, which indicates that both methods are accurate. While there are multiple partial matches (one haplotype is mixed according to a method and either sensitive or resistant by the other) with mixed haplotypes called from genomes, no partial match with mixed haplotypes called from barcodes were found, showing that whole genome sequencing is more sensitive than molecular genotyping.

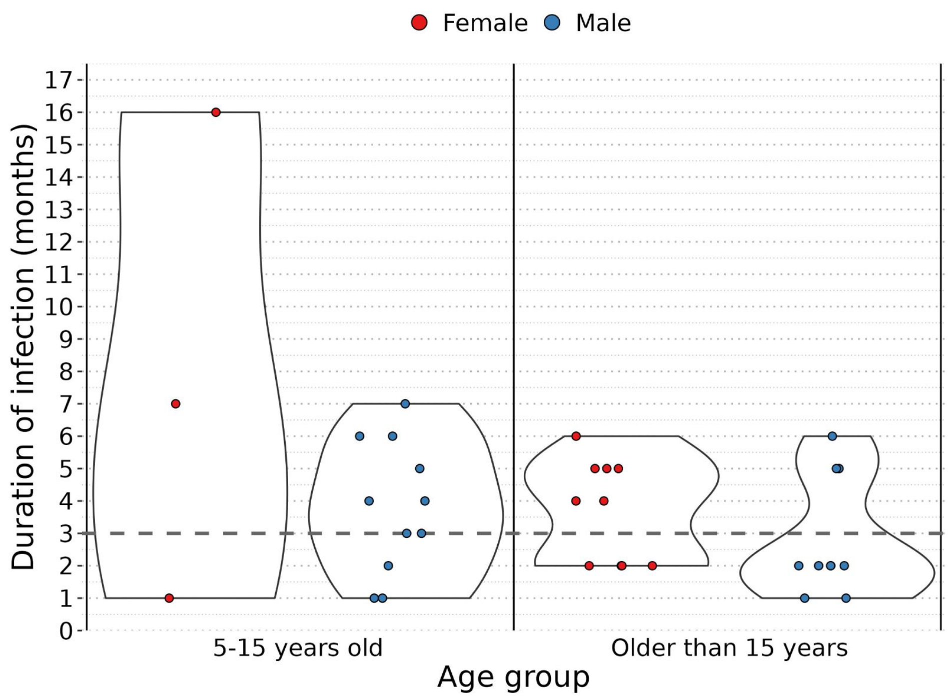


Figure S10

Durations of infection by the same parasite strain, grouped by host age and coloured by gender.

The number of short (less than 3 months) and long infections (more than 3 months) were compared between genders (female versus male) and age groups ('5-15 years old' versus 'older than 15 years'). No significant difference of the duration of infection was found between genders, which had very similar percentages of short infections with 38 % and 48 % of infections for female and male respectively (χ^2 = 0.30, p-value > 0.5). Although long infections were more common in the '5-15 years old' age group (71 % of all infections) compared to the 'older than 15 years age' group (45 % of all infections), this was not statistically significant (χ^2 = 2.34, p-value < 0.13), potentially due to relatively small sample size.

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Additional information

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Author contributions

Conceptualization by AC, TB, UDA and DC; field work and sample processing by SC, SD, FJ. Data analysis by MAG and AC. Writing of original draft by MAG and AC. Reviewing and editing of manuscript by all authors. Funding acquisition by AC and TB.

Additional files

Supplementary tables. *Table S1: Gender, age and households of participants. Table S2: Sampling, falciparum-malaria test results and treatment. Table S3: Locations of genotyped SNPs and their rank in barcodes. Table S4: Molecular barcode genotyping raw data and statistics. Table S5: Genome sequencing accession identifiers and statistics. Table S6: Complexity of infection metrics from barcodes. Table S7: Complexity of infection metrics from genomes. Table S8: Drug resistance markers genotyped from barcodes. Table S9: Drug resistance markers genotyped from genomes. Table S10: Duration of continuous infection estimated from barcode relatedness.*

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Reviewer #2 (Public review):

Summary:

Malaria transmission in the Gambia is highly seasonal, whereby periods of intense transmission at the beginning of the rainy season are interspersed by long periods of low to no transmission. This raises several questions about how this transmission pattern impacts the spatiotemporal distribution of circulating parasite strains, how parasites persist during the dry season, and how asymptomatic infections contribute to maintaining transmission during the low/no transmission season.

Combining a molecular barcode genotyping using 101 bi-allelic SNPs and SNPs from Whole Genome Sequence (WGS) in a "consensus barcode", the authors aimed at measuring the relatedness between parasites at different spatial (i.e., individual, household, village, and region) and temporal (i.e., high, low, and the corresponding the transitions) levels by assessing the fraction of the genome having a common ancestry (i.e. Identity-by-Descent (IBD)).

By measuring the Complexity of Infection (COI) and parasite relatedness by IBD the authors show that a large fraction of infections is polygenomic and stable over time, resulting in a high recombinational diversity. Moreover, they show that transmission intensity increases during the transition from the dry to wet seasons. However, they find that there is a higher probability of finding similar genotypes within the same household, but this similarity rapidly disappears over time and is not observed between different villages. If there is no drug selection during the dry season, and if resistance results in a fitness cost, alleles associated with drug resistance may change in frequency. The authors looked at the frequencies of six drug-resistance haplotypes (aat1, crt, dhfr, dhps, kelch13, and mdr1), and found no evidence of changes in allele frequencies associated with seasonality. They also find chronic infections lasting from one month to one and a half years with no dependence on age or gender.

This work makes use of genomic information and IBD analytic tools to show parasite relatedness from asymptomatic infections at different spatial and temporal scales, thus providing a better understanding of the transmission dynamics of malaria in highly seasonal environments.

Strength:

The authors use a combination of high-quality barcodes (425 barcodes representing 101 bi-allelic SNPs) and 199 high-quality genome sequences to infer the fraction of the genome with shared Identity by Descent (IBD) (i.e. a metric of recombination rate) over several time points covering two years. The barcode and whole genome sequence combination allows full use of a large dataset, to confidently infer the relatedness of parasite isolates at various spatiotemporal scales and show the advantage of using genomic information for understanding malaria transmission dynamics.

The authors aimed to establish how seasonal transmission cycles shape the spatiotemporal parasite population structure using metrics such as parasite genetic diversity, genetic relatedness, and frequency of drug resistance alleles, as well as the contribution of asymptomatic chronic carriers to sustained transmission. The results support their conclusions.

Using a combination of molecular barcodes and available whole genome sequence datasets opens new opportunities to understand malaria transmission dynamics in different transmission settings. This allows for data analysis at different spatiotemporal granularities, having a practical utility for identifying malaria control targets and acquiring metrics to evaluate malaria control programs. The development of molecular barcodes using similar SNPs by different malaria control programs would be of great utility to compare and understand malaria transmission dynamics in different settings worldwide.

https://doi.org/10.7554/eLife.103047.2.sa2

Reviewer #3 (Public review):

This study aimed to examine the impact of seasonality on the population genetics of malaria parasites. To achieve this, the researchers conducted a longitudinal study in a region with seasonal malaria transmission. Over a 2.5-year period, blood samples were collected from 1,516 participants across four villages in the Upper River Region of The Gambia. These samples were tested for malaria parasite infection, and the parasites from positive samples were genotyped using a genetic barcode and/or whole genome sequencing. Genetic relatedness analysis was then performed to explore the findings

The study identified three key findings:

(1) The malaria parasite population undergoes continuous recombination, with no single genotype predominating, in contrast to viral populations;

(2) Parasite relatedness is influenced by both spatial and temporal factors; and

(3) The lowest genetic relatedness among parasites occurs during the transition from the low to high transmission seasons, which the authors linked to increased recombination during sexual reproduction in mosquitoes.

The results section is well-structured, and the figures are clear and self-explanatory. The methods are adequately described, providing a solid foundation for the findings. While there are no unexpected results, it is reassuring to see the anticipated outcomes supported by actual data. The conclusions are generally well-supported and the recommendation to target asymptomatic infections is logical and relevant.

https://doi.org/10.7554/eLife.103047.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The manuscript titled "Household clustering and seasonal genetic variation of Plasmodium falciparum at the community-level in The Gambia" presents a valuable genetic spatio-temporal analysis of malaria-infected individuals from four villages in The Gambia, covering the period between December 2014 and May 2017. The majority of samples were analyzed using a SNP barcode with the Spotmalaria panel, with a subset validated through WGS. Identity-by-descent (IBD) was calculated as a measure of genetic relatedness and spatio-temporal patterns of the proportion of highly related infections were investigated. Related clusters were detected at the household level, but only within a short time period.*
>
> *Strengths:*
>
> *This study offers a valuable dataset, particularly due to its longitudinal design and the inclusion of asymptomatic cases. The laboratory analysis using the Spotmalaria platform combined and supplemented with WGS is solid, and the authors show a linear correlation between the IBD values determined with both methods, although other studies have reported that at least 200 SNPs are required for IBD analysis. Data-analysis pipelines were created for (1) variant filtering for WGS and subsequent IBD analysis, and (2) creating a consensus barcode from the spot malaria panel and WGS data and subsequent SNP filtering and IBD analysis.*
>
> *Weaknesses:*
>
> *Further refining the data could enhance its impact on both the scientific community and malaria control efforts in The Gambia.*
>
> *(1) The manuscript would benefit from improved clarity and better explanation of results to help readers follow more easily. Despite familiarity with genotyping, WGS, and IBD analysis, I found myself needing to reread sections. While the figures are generally clear and well-presented, the text could be more digestible. The aims and objectives need clearer articulation, especially regarding the rationale for using both SNP barcode and WGS (is it to validate the approach with the barcode, or is it to have less missing data?). In several analyses, the purpose is not immediately obvious and could be clarified.*

The text of the manuscript has now been thoroughly revised. But please let us know if a specific section remains unclear.

> *(2) Some key results are only mentioned briefly in the text without corresponding figures or tables in the main manuscript, referring only to supplementary figures, which are usually meant for additional detail, but not main results. For example, data on drug resistance markers should be included in a table or figure in the main manuscript.*

We agree with the reviewer suggesting to move the prevalence of drug resistance markers from supplementary figures (previously Figure S8) to the main manuscript (now Figure 5). If other Figure/Table should be moved to the main manuscript please let us know.

> *(3) The study uses samples from 2 different studies. While these are conducted in the same villages, their study design is not the same, which should be addressed in the*

> *interpretation and discussion of the results. Between Dec 2014 and Sept 2016, sampling was conducted only in 2 villages and at less frequent intervals than between Oct 2016 to May 2017. The authors should assess how this might have impacted their temporal analysis and conclusions drawn. In addition, it should be clarified why and for exactly in which analysis the samples from Dec 2016 - May 2017 were excluded as this is a large proportion of your samples.*

We have clarified which set of samples was used in our Results (Lines 293-295, 316-319). While two villages were recruited halfway through the study, two villages (J and K, Figure 1C) consistently provided data for each transmission season. Importantly, our temporal analysis accounts for these differences by grouping paired barcodes based on their respective locations (Figure 3B). Despite variations in sampling frequency, we still observe a clear overall decline in relatedness between the ‘0-2 months’ and ‘2-5 months’ groups, both of which include barcodes from all four villages.

> *(4) Based on which criteria were samples selected for WGS? Did the spatiotemporal spread of the WGS samples match the rest of the genotyped samples? I.e. were random samples selected from all times and places, or was it samples from specific times/places selected for WGS?*

All *P. falciparum* positive samples were sent for genotyping and whole genome sequencing, ensuring no selection bias. However, only samples with sufficient parasite DNA were successfully sequenced. We have updated the text (Line 129-130) and added a supplementary figure (Figure S4) to show the sample collection broken down by type of data (barcode or genome). High quality genomes are distributed across all time points.

> *(5) The manuscript would benefit from additional detail in the methods section.*

Please see our response in the section “Recommendation for the authors”.

> *(6) Since the authors only do the genotype replacement and build consensus barcode for 199 samples, there is a bias between the samples with consensus barcode and those with only the genotyping barcode. How did this impact the analysis?*

While we acknowledge the potential for bias between samples with a consensus barcode (based on WGS) and those with genotyping-only barcodes, its impact is minimal. WGS does indeed produce a more accurate barcode compared to SNP genotyping, but any errors in the genotyping barcodes were mitigated by excluding loci that systematically mismatched with WGS data (see Figure S3). Additionally, the use of WGS improved the accuracy of 51 % (216/425) of barcodes, which strengthens the overall quality and validity of our analysis.

> *(7) The linear correlation between IBD-values of barcode vs genome is clear. However, since you do not use absolute values of IBD, but a classification of related (>=0.5 IBD) vs. unrelated (<0.5), it would be good to assess the agreement of this classification between the 2 barcodes. In Figure S6 there seem to be quite some samples that would be classified as unrelated by the consensus barcode, while they have IBD>0.5 in the Genome-IBD; in other words, the barcode seems to be underestimating relatedness.*
>
> *a. How sensitive is this correlation to the nr of SNPs in the barcode?*

We measured the agreement between the two classifications using specificity (0.997), sensitivity (0.841) and precision (0.843) described in the legend of Figure S8. To further demonstrate the good agreement between the two methods, we calculated a Cohen’s kappa value of 0.839 (Lines 226, 290), indicative of a strong agreement (McHugh 2012). As expected, the correlation between IBD values obtained by both methods improves (higher Cohen's kappa and R^2) as the cutoff for the minimal number of comparable and informative loci per barcode pair is raised (data not shown).

> *(8) With the sole focus on IBD, a measure of genetic relatedness, some of the conclusions from the results are speculative.*
>
> *a. Why not include other measures such as genetic diversity, which relates to allele frequency analysis at the population level (using, for example, nucleotide diversity)? IBD and the proportion of highly related pairs are not a measure of genetic diversity. Please revise the manuscript and figures accordingly.*

We agree with the fact that IBD is not a direct measure of genetic diversity, even though both are related (Camponovo et al., 2023). More precisely, IBD is a measure of the level of inbreeding in the population (Taylor et al., 2019). We have updated our manuscript by replacing "genetic diversity" with "genetic relatedness" or "inbreeding/outcrossing" when appropriate. Nucleotide diversity would be relevant if we wanted to compare different settings, e.g. Africa vs Asia, however this is not the case here.

> *b. Additionally, define what you mean by "recombinatorial genetic diversity" and explain how it relates to IBD and individual-level relatedness.*

We considered the term 'recombinatorial genetic diversity' to be equivalent to the level of inbreeding in the population. Because this expression is rather uncommon, we decided to drop it from our manuscript and replace it with "inbreeding/outcrossing".

> *c. Recombination is one potential factor contributing to the loss of relatedness over time. There are several other factors that could contribute, such as mobility/gene flow, or study-specific limitations such as low numbers of samples in the low transmission season and many months apart from the high transmission samples.*

Indeed, the loss of relatedness could be attributed not only to the recombination of local cases but also to new parasites introduced by imported malaria cases. As we stated in our manuscript, previous studies have shown a limited effect of imported cases on maintaining transmission (Lines 72-74). Nevertheless, we cannot definitely exclude that imported cases have an effect on inbreeding levels, since we do not have access to genetic data of surrounding parasites at the time of the study. We updated the discussion accordingly (Lines 497-501).

> *d. By including other measures such as linkage disequilibrium you could further support the statements related to recombination driving the loss of relatedness.*

This commendable suggestion is actually part of an ongoing project focusing on the sharing of IBD fragments and how it correlates with linkage disequilibrium. However, we believe that this analysis would not fit in the scope of our manuscript which is really about spatio-temporal effects on parasite relatedness at a local scale.

> *(9) While the authors conclude there is no seasonal pattern in the drug-resistant markers, one can observe a big fluctuation in the dhps haplotypes, which go down from 75% to 20% and then up and down again later. The authors should investigate this in more detail, as dhps is related to SP resistance, which could be important for seasonal malaria chemoprofylaxis, especially since the mutations in dhfr seem near-fixed in the population, indicating high levels of SP resistance at some of the time points.*

As the reviewer noted, the DHPS A437G haplotype appears to decrease in prevalence twice throughout our study: from the 2015 and 2016 high transmission seasons to the subsequent 2016 and 2017 low transmission seasons. Seasonal Malaria Chemoprophylaxis (SMC) was carried out in the area through the delivery of sulfadoxine–pyrimethamine plus amodiaquine to children 5 years old and younger during high transmission seasons. As DHPS A437G haplotype has been associated with resistance to sulfadoxine, its apparent increase in prevalence during high transmission seasons could be resulting from the selective pressure imposed on parasites. After SMC, the decrease in prevalence observed during low transmission seasons could be caused by a fitness cost of the mutation favouring wild-type parasites over resistant ones. We updated our manuscript to reflect this relevant observation (Lines 400-405).

> *(10) I recommend that raw data from genotyping and WGS should be deposited in a public repository.*

Genotyping data is available in the supplementary table 4 (Table S4). Whole genome sequencing is accessible in a European Nucleotide Archive public repository with the identifiers provided in supplementary table 5 (Table S5). We added references to these tables in the manuscript (Lines 249-250).

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Malaria transmission in the Gambia is highly seasonal, whereby periods of intense transmission at the beginning of the rainy season are interspersed by long periods of low to no transmission. This raises several questions about how this transmission pattern impacts the spatiotemporal distribution of circulating parasite strains. Knowledge of these dynamics may allow the identification of key units for targeted control strategies, the evaluation of the effect of selection/drift on parasite phenotypes (e.g., the emergence or loss of drug resistance genotypes), and analyze, through the parasites' genetic nature, the duration of chronic infections persisting during the dry season. Using a combination of barcodes and whole genome analysis, the authors try to answer these questions by making clever use of the different recombination rates, as measured through the proportion of genomes with identity-by-descent (IBD), to investigate the spatiotemporal relatedness of parasite strains at different spatial (i.e., individual, household, village, and region) and temporal (i.e., high, low, and the corresponding the transitions) levels. The authors show that a large fraction of infections are polygenomic and stable over time, resulting in high recombinational diversity (Figure 2). Since the number of recombination events is expected to increase with time or with the number of mosquito bites, IBD allows them to investigate the connectivity between spatial levels and to measure the fraction of effective recombinational events over time. The authors demonstrate the epidemiological connectivity between villages by showing the presence of related genotypes, a higher probability of finding similar genotypes within the same household, and how parasite-relatedness gradually disappears over time (Figure 3). Moreover, they show that transmission intensity increases during the transition from dry to wet seasons (Figure 4). If there is no drug selection during the dry season and if resistance incurs a fitness cost it is possible that alleles associated with drug resistance may change in frequency. The authors looked at the frequencies of six drug-resistance haplotypes (aat1, crt, dhfr, dhps, kelch13, and mdr1), and found no evidence of changes in allele frequencies associated with seasonality. They also find chronic infections lasting from one month to one and a half years with no dependence on age or gender.*

The use of genomic information and IBD analytic tools provides the Control Program with important metrics for malaria control policies, for example, identifying target populations for malaria control and evaluation of malaria control programs.

Strength:

The authors use a combination of high-quality barcodes (425 barcodes representing 101 bi-allelic SNPs) and 199 high-quality genome sequences to infer the fraction of the genome with shared Identity by Descent (IBD) (i.e. a metric of recombination rate) over several time points covering two years. The barcode and whole genome sequence combination allows full use of a large dataset, and to confidently infer the relatedness of parasite isolates at various spatiotemporal scales.

Reviewer #3 (Public review):

Summary

This study aimed to investigate the impact of seasonality on the malaria parasite population genetic. To achieve this, the researchers conducted a longitudinal study in a region characterized by seasonal malaria transmission. Over a 2.5-year period, blood samples were collected from 1,516 participants residing in four villages in the Upper River Region of The Gambia and tested the samples for malaria parasite positivity. The parasites from the positive samples were genotyped using a genetic barcode and/or whole genome sequencing, followed by a genetic relatedness analysis.

The study identified three key findings:

(1) The parasite population continuously recombines, with no single genotype dominating, in contrast to viral populations;

(2) The relatedness of parasites is influenced by both spatial and temporal distances; and

(3) The lowest genetic relatedness among parasites occurs during the transition from low to high transmission seasons. The authors suggest that this latter finding reflects the increased recombination associated with sexual reproduction in mosquitoes.

The results section is well-structured, and the figures are clear and self-explanatory. The methods are adequately described, providing a solid foundation for the findings. While there are no unexpected results, it is reassuring to see the anticipated outcomes supported by actual data. The conclusions are generally well-supported; however, the discussion on the burden of asymptomatic infections falls outside the scope of the data, as no specific analysis was conducted on this aspect and was not stated as part of the aims of the study. Nonetheless, the recommendation to target asymptomatic infections is logical and relevant.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The manuscript would benefit from additional detail in the methods section.

a. Refer to Figure 1 when you describe the included studies and sample processing.

We added the reference to Figure 1 (Line 131).

b. While you describe each step in the pipeline, you do not specify the tools, packages, or environment used (the GitHub link is also non-functional). A graphic representation of

> *the pipeline, with more bioinformatic details than Supplementary Figure S1, would be helpful. Add references to used tools and software created by others.*

The GitHub link has been updated and is now functional. We find Figure S1 already heavy in details, adding in more would be detrimental to our will of it being an easily readable summary of our pipeline. Readers seeking in-depth explanation of our pipeline might be more interested in reading the methods section instead. We are very much committed to credit the authors of the tools that were essential for us to create our analysis pipeline. The two most relevant tools that we used are hmmIBD and the Fws calculation, which were both cited in the methods (Lines 148-152, 214-215).

> *c. What changed in the genotyping protocol after May 2016? Does it not lead to bias in the (temporal) analysis by leaving these loci in for samples collected before May 2016 and making them 'unknown' for the majority of samples collected after this date?*

These 21 SNPs all clustered in 1 of the 4 multiplexes used for molecular genotyping, which likely failed to produce accurate base calls. We updated the text to include this information (Lines 198-200).

The rationale behind the discarding of these 21 SNPs for barcodes sampled after May 2016 was that they were consistently mismatching with the WGS SNPs, probably due to genotyping error as mentioned above. However, by replacing these unknown positions in the molecular barcodes with WGS SNPs, 141 samples did recover some of these 21 SNPs with the accurate base calls (Figure S3A). Additionally, we added an extra analysis to assess the agreement between barcodes and WGS data (Figure S3B).

> *d. Related to this, how are unknown and mixed genotypes treated in the binary matrix? How is the binary matrix coded? Is 0 the same as the reference allele? So all the missing and mixed are treated as references? How many missing and mixed alleles are there, how often does it occur and how does this impact the IBD analysis?*

We acknowledge that the details that we provided regarding the IBD analysis were confusing. hmmIBD requires a matrix that contains positive or null integers for each different allele at a given loci (all our loci were bi-allelic, thus only 0 and 1 were used) and -1 for missing data. In our case, we set missing and mixed alleles to -1, which were then ignored during the IBD estimation. The corresponding text was updated accordingly (Lines 173-175).

> *e. By excluding households with less than 5 comparisons, are you not preselecting households with high numbers of cases, and therefore higher likelihood of transmission within the household?*

All participants in each household were sampled at every collection time point. This sampling was unbiased towards likelihood of transmission. Excluding pairs of households with less than 5 comparisons was necessary to ensure statistical robustness in our analyses. Besides, this does not necessarily restrict the analysis to only households with a high number of cases as it is the total number of pairs between households that must equal 5 at least (for instance these pairs would pass the cutoff: household with 1 case vs household with 5 cases; household with 2 cases vs household with 3 cases).

> *(2) Since the authors only do the genotype replacement and build consensus barcode for 199 samples, there is a bias between the samples with consensus barcode and those with only the genotyping barcode. How did this impact the analysis?*

See (6) in the Public Review.

> *a. It would be good to get a better sense of the distribution of the nr of SNPs in the barcode. The range is 30-89, and 30 SNPs for IBD is really not that much.*

Adding the range of the number of available SNPs per barcode is indeed particularly relevant. We added a supplementary figure (Figure S5) showing the distribution of homozygous SNPs per barcode, showing that a very small minority of barcodes have only 30 SNPs available for IBD (average of 65, median of 64).

> *b. Did you compare the nr of SNPs in the consensus vs. only genotyped barcodes? Is there more missing data in the genotype-only barcodes?*

We added a supplementary figure (Figure S5) with the distribution of homozygous SNPs in consensus (216 samples) and molecular (209 samples) barcodes. Consensus barcodes have more homozygous SNPs (average 76, median 82) than molecular barcodes (average of 54, median of 53), showing the improvement resulting from using whole genome sequencing data.

> *c. How was the cut-off/sample exclusion criteria of 30 SNPs in the barcode determined?*

As described above (Public review section 7.a.), we removed pairs of barcodes with less than 30 comparable loci (and 10 informative loci) because this led to a good agreement between IBD values obtained from barcodes and genomes while still retaining a majority of pairwise IBD values.

> *d. Was there more/less IBD between sample pairs with a consensus barcode vs those with genotype-only barcodes?*

We separated pairwise IBD values into two groups: "within consensus" and "within molecular". The percentages of related barcodes (IBD ≥ 0.5) was virtually identical between "within consensus" (1.88 %) and "within molecular" (1.71 %) groups (χ^2 = 1.33, p value > 0.24).

> *(3) Line 124 adds a reference for the PCR method used.*

We have updated this information: varATS qPCR (Line 121).

> *(4) Line 126, what is MN2100ff? Is this the catalogue number of the cellulose columns? Please clarify and add manufacturer details.*

MN2100ff was a replacement for CF11. We added a link to the MalariaGen website describing the product and the procedure (Lines 124-125).

> *(5) Line 143: Figure S7 is the first supplementary figure referenced. Change the order and make this Figure S1?*

The numbering of figures is now fixed.

> *(6) Line 154: How many SNPs were in the vcf before filtering?*

There were 1,042,186 SNPs before filtering. This information was added to the methods (Line 168).

> *(7) Line 156: Why is QUAL filtered at 10000? This seems extremely high. (I could be mistaken, but often QUAL above 50 or so is already fine, why discard everything below*

> *10000?). What is the range of QUAL scores in your vcf?*

We used the QUAL > 10000 to make our analyses less computationally intensive while keeping enough relevant genetic information. We agree that keeping variants with extremely high values of QUAL is not relevant above a certain threshold as it translates into infinitesimally low probabilities ($10^{-(QUAL/10)}$) of the variant calling being wrong. We then decided to use a minimal population minor allele frequency (MAF) of 0.01 to keep a variant as this will make the IBD calculation more accurate (Taylor et al., 2019). The variant filtering was carried out with the MAF > 0.01 filter, resulting in 27,577 filtered SNPs with a minimal QUAL of 132. With a cutoff of 3000 available SNPs, we retrieved all 199 genomes previously obtained with the QUAL > 10000 condition. The methods have been updated accordingly (Lines 166-170).

> *(8) Line 161-165: How did you handle the mixed alleles in the hmmIBD analysis for the WGS data? Did you set them as 0 as you do later on for the consensus barcode?*

Mixed alleles and missing data were ignored. This translated into a value of -1 for the hmmIBD matrix and not 0 as we incorrectly stated previously. We updated our manuscript with this correct information (Lines 173-175).

> *(9) Line 168-171: How many SNPs do you have in the WGS dataset after all the filtering steps? If the aim of the IBD with WGS was to validate the IBD-analysis with the barcode, wouldn't it make sense to have at least 200 loci (as shown in Taylor et al to be required for hmmIBD) in the WGS data? What proportion of comparisons were there with only 100 pairs of loci? This seems like really few SNPs from WGS data.*

There were 27,577 SNPs overall in the 199 high quality genomes. In our analysis, we make the distinction between comparable and informative loci. For two loci to be comparable, they both have to be homozygous. To be informative, they must be comparable and at least one of them must correspond to the minor allele in the population. We borrowed this term and definition from hmmIBD software which yields directly the number of informative loci per pair. By keeping pairs with at least 100 informative SNPs, we aimed to reduce the number of samples artificially related because only population major alleles are being compared. Pairs of genomes had between 1073 and 27466 of these, way above the recommended 200 loci in Taylor et al. (2019). We added more details on comparable and informative sites (Lines 152-160).

> *(10) Line 178: why remove the 12 loci that are absent from the WGS? Are these loci also poorly genotyped in the spotmalaria panel?*

As our goal is to validate the reliability of molecular genotyped SNPs, these 12 loci have to be removed. Especially because we did find a consistent discrepancy between genotyped and WGSed SNPs, which cannot be tested if these SNPs are absent from the genomes.

> *(11) Line 180-182: What do you mean by this sentence: "Genomic barcodes are built using different cutoffs of within-sample MAF and aligned against molecular barcodes from the same isolates." Is this the analysis presented in the supplementary figure and resulting in the cut-off of MAF 0.2? Please clarify.*

A loci where both alleles are called can result from two distinct haploïd genomes present or from an error occurring during sequencing data acquisition or processing. To distinguish between the two, we empirically determined the cutoff of within-sample MAF above which the loci can be considered heterozygous and below which only the major allele is kept. The corresponding figure was indeed Figure S2 (referenced in next sentence Lines 192-195). We clarified our approach in the methods (Lines 190-192) and legends of Figures S2 and Figure S3.

> *(12) Line 191: How often was there a mismatch between WGS and SNP barcode?*

We added a panel (Figure S3B) showing the average agreement of each SNP between molecular genotyping and WGS. We highlighted the 21 discrepant SNPs showing a lower agreement only for samples collected after May 2016.

> *(13) Line 201-204: This part is unclear (as above for the WGS): did you include sample pairs with more than 10 paired loci? But isn't 10 loci way too few to do IBD analysis?*

We included pairs of samples with at least 30 comparable loci and 10 informative paired loci (refer to our answer to comment 8 for the difference between the two). We added more details regarding comparable and informative sites (Lines 152-160). Indeed, using fewer than 200 loci leads to an IBD estimation that is on average off by 0.1 or more (Taylor et al., 2019). However we showed that the barcode relatedness classification based on a cutoff of IBD (related when above 0.5, unrelated otherwise) was close enough to our gold standard using genomes (each pair having more than 1000 comparable sites). Because we use this classification approach rather than the exact value of barcode-estimated IBD in our study, our 30 minimum comparable sites cutoff seems sufficient.

> *(14) Lines 206-207: which program did you use to analyse Fws?*

We did not use any program, we computed Fws according to Manske et al. (2012) methods.

> *(15) Line 233: "we attempted parasite genotyping and whole genome sequencing of 522 isolates over 16 time points" => This is confusing, you did not do WGS of 522 samples, only 199 as mentioned in the next sentence.*

We attempted whole genome sequencing on 331 isolates and molecular genotyping on 442 isolates with 251 isolates common between the two methods. We updated our text to clarify this point (Lines 247-252).

> *(16) Lines 256-259: Add a range of proportions or some other summary statistic in this section as you are only referring here to supplementary figures to support these statements.*

The text has been updated (Lines 271-274).

> *(17) Line 260: check the formatting of the reference "Collins22" as the rest of the document references are numbered.*

Fixed.

> *(18) Figure 2/3:*
>
> *a. You could also inspect relatedness at the temporal level, by adjusting the network figure where the color is village and shape is time (month/year).*

Although visualising the effect of time on the parasite relatedness network would be a valuable addition, we did not find any intuitive and simple way of doing so. Using shapes to represent time might end up being more confusing than helpful, especially because the sampling was not done at fixed intervals.

> *b. To further support the statement of clustering at the household level, it might be useful to add a (supplementary) figure with the network with household number/IDs as color or shape. In the network, there seems to be a lot of relatedness within the villages and between villages. Perhaps looking only at the distribution of the proportion of highly related isolates is simplifying the data too much. Besides, there is no statistical difference between clustering at the household vs within-village levels as indicated in Figure 3.*

Unfortunately, there are too many households (71 in Figure 2) to make a figure with one color or shape per household readable. The statistical test of the difference between the within household and within village relatedness yielded a p value above the cutoff of 0.05 (p value of 0.084). However, it is possible that the lack of significance arises from the relatively low number of data points available in the "within household" group. This is even more plausible considering the statistical difference of both "within household" and "within village" groups with "between village" group. Overall, our results indicate a decreasing parasite relatedness with spatial distance, and that more investigation would be needed to quantify the difference between "within household" and "within village" groups.

> *(19) Figure 4: Please add more description in the caption of this figure to help interpret what is displayed here. Figure 4A is hard to interpret and does not seem to show more than is already shown in Figure 3A. What do the dots represent in Figure 4B? It is not clear what is presented here.*

Compared to Figure 3A, Figure 4A enables the visualization of the relatedness between each individual pair of time points, which are later used in the comparison of relatedness between seasonal groups in Figure 4B. For this reason, we believe that Figure 4A should remain in the manuscript. However, we agree that the relationship between Figure 4A and Figure 4B is not intuitive in the way we presented it initially. For this reason, we added more details in the legend and modified Figure 4A to highlight the seasonal groups used in Figure 4B.

> *(20) Line 360-361: what did you do when haplotypes were not identical?*

We explained it in the methods section (Lines 144-146): in this case, only WGS haplotypes were kept.

> *(21) Section chronic infections: it is important to mention that the majority of chronic infections are individuals from the monthly dry-season cohort.*

We added a statement about the 21 chronically infected individuals that were also part of the December 2016 – May 2017 monthly follow-up (Lines 423-426).

> *(22) Lines 381-386: Did you investigate COI in these individuals? Could it be co-circulating strains that you do not pick up at all times due to the consensus barcodes and discarding of mixed genotypes (and does not necessarily show intra-host competition. That is speculation and should perhaps not be in the results)?*

This is exactly what we think is happening. Due to the very nature of genotyping, only one strain may be observed at a time in the case of a co-infection, where distinct but related strains are simultaneously present in the host. The picked-up strain is typically the one with the highest relative abundance at the time of sampling. As the reviewer stated, fluctuation of strain abundance might not only be due to intra-host competition but also asynchronous development stages of the two strains. We added this observation to the manuscript (Lines 432-435).

> *(22) Figure 6: highlight the samples where the barcode was not available in a different color to be able to see the difference between a non-matching barcode and missing data.*

We thank the reviewer for this great suggestion. We have now added to Figure 6 barcodes available along with their level of relatedness with the dominant genotypes for each continuous infections.

> *(24) Improve the discussion by adding a clear summary of the main findings and their implications, as well as study-specific limitations.*

The Discussion has been updated with a paragraph summarizing the primary results (Lines 451-457).

> *(25) Line 445: "implying that the whole population had been replaced in just one year "*
>
> *a. What do you mean by replaced? Did other populations replace the existing populations? I am not sure the lack of IBD is enough to show that the population changed/was replaced. Perhaps it is more accurate to say that the same population evolved. Nevertheless, other measures such as genetic diversity and genetic differentiation or population structure.would be more suitable to strengthen these conclusions.*

We agree that "replaced" was the wrong term in this case. We rather intended to describe how the numerous recombinations between malaria parasites completely reshaped the same initial population which gradually displayed lower levels of relatedness over time. We updated the manuscript accordingly (Lines 507-512).

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) Line 260: Remove Collins 22.*

Fixed.

> *(2) Lines 270-274: 73 + 213 = 286 not 284; sum of percentages is equal to 101%.*

The numbers are correct: the 73 barcodes identical (IBD >= 0.9) to another barcode are a subset of the 213 related (IBD >= 0.5) to another barcode. However we agree that this might be confusing and will considering barcodes to be related if they have an IBD between 0.5 and 0.9, while excluding those with an IBD >= 0.9. The text has been updated (Lines 299-301).

> *(3) Section: "Independence of seasonality and drug resistance markers prevalence".*

The text has been revised and the supplementary figure is now a main figure.

> *(4) For readers unaware of malaria control policy in the Gambia it would be helpful to have more details on the specifics of anti-malarial drug administration.*

We added the drugs used in SMC (sulfadoxine-pyrimethamine and amodiaquine) and the first line antimalarial treatment in use in The Gambia during our study (Coartem) (Lines 383-388).

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) The abstract is not as clear as the authors' summary. For example, I found the sentence starting with "with 425 P. falciparum..." hard to follow.*

The abstract has been updated.

> *(2) It is better to consistently use "barcode genotyping "or "genotyping by barcode". Sometimes "molecular genotyping" is used instead of "barcode genotyping"*

We have now replaced all occurrences of “barcode genotyping” with “molecular genotyping” or “molecular barcode genotyping”. We prefer to stick with “molecular genotyping” as this let us distinguish between the molecular and the genomic barcode.

> *(3) The introduction is quite disjoined and does not provide a clear build-up to the gap in knowledge that the study is attempting to fill. please revise.*

Introduction is now thoroughly revised.

> *(4) Line 31 "with notable increase of parasite differentiation" is an interpretation and not an observation.*

We have modified that sentence (Lines 31-33).

> *(5) Overall, the introduction requires substantial revision.*

Introduction is now thoroughly revised.

> *(6) Line 70 "parasite population adapts..." I thought this required phenotypic analysis and not genetics?*

The idea is that population of parasites may adapt to environmental conditions (such as seasonality) by selecting the most fitted genotypes. For instance, antimalarial exposure has an effect of selecting parasites with specific mutations in drug resistance related genes, and this even appears to be transient (for example with chloroquine). As such, there is good reason to think that seasonality might have a similar effect on parasite genetics.

> *(7) Line 129-130: the #442 is not reflected in the schematic Figure 1.*

This is an intentional choice to make the figure more synthetic. For this reason, we included the Figure S1, which provides more details on the data collection and analysis pipeline.

> *(8) Line 242-243: "Made with natural earth". What is this?*

This is a statement acknowledging the use of Natural Earth data to produce the map presented in Figure 1A.

> *(9) Line 260: "collins22", is this a reference?*

Fixed.

> *(10) Line 269-70. Very hard to follow. Please revise.*

We changed the text (Lines 293-297).

> *(11) Line 324: similarly... I think there is a typo here.*

We did not find any typo in this specific sentence. However, “Similarly to Figure 3” sounds maybe a bit off, so we changed it to “As in Figure 3” (Line 351).

> *(12) Line 332-334: very hard to follow. please revise. Again, the lower parasite relatedness during the transition from low to high was linked to recombination occurring in the mosquito but what about infection burden shifting to naive young children? Is there a role for host immunity in the observed reduction in parasite-relatedness during the transition period?*

This text has been rewritten (Lines 356-361).

About the hypothesis of infection burden shifting to naïve young children, this question is difficult to address in The Gambia because children under 5 years old received Seasonal Malaria Chemoprophylaxis during the high transmission season. In older children (6-15 years old), the prevalence was similar to adults (Fogang et al., 2024).

About the role of host immunity on parasite relatedness across time and space, our dataset is too small to divide it in different age groups. Further studies should address this very interesting question.

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